JHS Journal *of* Historical Syntax
Volume 10, Article 5: 1–79, 2026 · DOI: 10.18148/hs/2026.v10i5.292

(NOT) ANYTHING GOES: THE SYNTAX AND DIACHRONY OF THE GERMAN *GEHEN*-MIDDLE*

Benjamin L. Sluckin
Ruhr-Universität Bochum

Eric Fuss
Ruhr-Universität Bochum

Abstract This paper offers a syntactic and diachronic analysis of the German [‘gehen + zu’-infinitive] construction. Previously misclassified as a periphrastic anticausative or conflated with canonical modal passives, the configuration represents a structurally distinct periphrastic realization of the middle voice. Synchronically, modal ‘gehen’ operates as a functional raising head that strictly selects a voiceless, eventive vP and a resultative configuration — a selectional restriction that directly accounts for the construction’s rigid lexical-semantic preference for Accomplishments. The raising head carries a formal edge feature forcing internal argument raising, which feeds dispositional evaluation at the conceptual-intentional interface. The resulting feature valuation simultaneously bleeds temporal auxiliary insertion and triggers post-syntactic ‘zu’-insertion as a morphotactic repair. Historical corpus data spanning the 15th to 20th centuries show that the ‘gehen’-middle emerged rapidly across the 18th and 19th centuries via a two-stage grammaticalization pathway: the lexical motion verb first underwent a semantic shift toward possibility, extending analogically into clausal control infinitives; ambiguous control structures were then reanalysed as monoclausal raising configurations. This reanalysis stripped the infinitival complement of its clausal architecture, reducing it to a bare vP, and bleached the matrix verb of its thematic subject, producing the constrained functional raising head attested in Present-Day German.

Keywords: middles; grammaticalization; dispositions; unaccusativity; raising; control; gehen.

* We thank audiences at DGfS 2023, DiGS 2025, and talks in Hamburg, Cologne and Berlin, as well as all friends, family, and colleagues confronted with countless judgments. We also thank Florian Schäfer for his useful comments on the paper. Finally, we thank the *Hausmeister* ‘Janitor’ for using this expression at exactly the right time, and the communal bin that, in his words, *nicht aufzuschließen ging*, i.e. wouldn’t unlock.

1 Introduction

German exhibits a set of constructions in which an apparently active infinitival complement after *zu* is used to express passive voice. Among the most widely discussed of these are modal passives formed with the copula *sein* ('to be'), which convey possibility or obligation (see Höhle 1978, Holl 2010, Abraham 2020). Far less understood is a related construction that combines a finite form of the lexical motion verb *gehen* ('to go') with a *zu*-infinitive. This configuration, illustrated in (1), involves the promotion of the internal argument to the subject position, where it receives nominative case, while the external argument is demoted.

(1) *Die Tür geht leicht aufzuschließen.*
the door goes easy open-to-close
'The door unlocks easily | the door is easy to unlock.'

Crucially, the construction imposes a dispositional property onto this promoted subject, yielding a strong modal interpretation of dynamic (im)possibility. In this respect, the configuration closely resembles a canonical Germanic middle. However, it conspicuously lacks the reflexive morphology (e.g., *sich* 'self') required to license middles in German.

To understand this construction, we must distinguish this monoclausal raising structure from a superficially similar biclausal control structure. As shown in (2), *gehen* can also operate as a semi-modal control verb evaluating a clausal subject.

(2) [*Tabus abzuschaffen*] *geht gar nicht so einfach.*
taboos off-to-abolish goes at.all not so easy
'It is really not that easy to abolish taboos.'
(Der Tagesspiegel, 30.11.1999, DWDS)

In the control configuration, the internal argument (*Tabus*) remains trapped within the embedded infinitival clause and retains accusative case. Furthermore, there is no agreement between the plural internal argument and the singular matrix verb *geht*. The monoclausal raising configuration in (1), by contrast, requires strict structural coherence and categorically bars the projection of an external argument.

The *gehen*-periphrasis has evaded rigorous empirical and theoretical investigation to date. Previous synchronic treatments have either relegated the construction to the margins of modal passives (Holl 2010, Müller 2019, Abraham 2020) or misclassified it entirely as an *Unmöglichkeitsantikausativ* ('impossibility anticausative') inherently driven by negation (Cysouw 2023).

This investigation provides, to our knowledge, the first extensive syntactic and diachronic analysis of the [*gehen*+*zu*-INFINITIVE] configuration. To capture its modern distribution, we deploy a formal acceptability judgment task, demonstrating that *gehen* operates as a functional raising head that enforces detransitivization through unaccusative syntax. Raising *gehen* strictly selects a voiceless, eventive *v*P and a resultative configuration. It then forces the internal argument to raise via a purely formal edge feature, which feeds a dispositional evaluation at the conceptual-intentional interface. Simultaneously, its prevalued status for [T] bleeds the insertion of temporal auxiliaries.

We argue that [*gehen*+*zu*-INFINITIVE] represents the direct structural reflex of a specific diachronic trajectory. By tracing the construction's emergence between the eighteenth and nineteenth centuries, we map a formal grammaticalization pathway wherein a semantic shift expressing possibility triggered the analogical extension of *gehen* into clausal control structures. These structures were subsequently reanalysed as the highly constrained monoclausal raising configurations observed today.

Section 2 establishes the theoretical and empirical background for *gehen*, detransitivization, and *zu*-infinitives. Section 3 presents the methodology and results of our quantitative acceptability judgment task. Section 4 discusses these empirical findings in theoretical terms. Section 5 provides a step-by-step formal derivation of the modern raising configuration. Finally, Section 6 contains a diachronic investigation making use of historical corpus data to propose a timeline for the reanalysis of [*gehen*+*zu*-INFINITIVE] from a control to a raising structure. We conclude the findings in Section 7.

2 Locating *gehen*: background and diagnostics

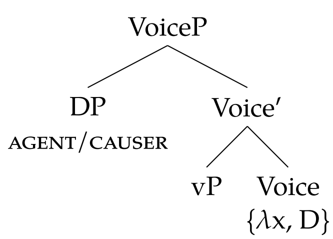
[*gehen*+*zu*-INFINITIVE] is difficult to classify. It shows argument-structural effects associated with both passives and middles without displaying the morphology of either, and it structurally resembles modal *zu*-infinitival passives (Haiden 2005, Holl 2010, Haider 2010) as well as predicates involving raising and control (Wurmbrand 2001, 2004, 2015, Haider 2010). Resolving its status therefore requires a battery of formal diagnostics. This section assembles only those diagnostics, and no more. We first fix a structural framework for Voice (§2.1); we then establish the behaviour of canonical passives and anticausatives (§2.2), the middle voice (§2.3), and modal infinitives and restructuring (§2.4). §2.5 applies this map to [*gehen*+*zu*-INFINITIVE], where the diagnostics converge on two competing predictions that the experiment in Section 3 is designed to decide between.

2.1 *A structural framework*

To evaluate [*gehen*+*zu*-INFINITIVE] against these strategies, we adopt the typology of Voice in Alexiadou, Anagnostopoulou & Schäfer (2015), on which transitivity alternations follow from two properties of the Voice head: whether it introduces an Agent or Causer (which we write as a λ-bound role, λx) and whether it bears a formal [D] feature forcing projection of a specifier. The four combinations relevant for German are shown in (3). **Active thematic Voice** {λx, D} merges the external argument in Spec,VoiceP, deriving transitives and unergatives. **Thematic non-active Voice** {λx, Ø} retains the Agent but lacks [D]: with no specifier the Agent is syntactically demoted yet remains active, deriving the *werden*-passive and licensing the agentive *von*-phrase. **Expletive active Voice** {Ø, D} bears [D] but no Agent: the expletive *sich* fills its specifier, deriving German's syntactically identical reflexive(-marked) anticausatives and canonical middles. A **voiceless** structure lacks VoiceP altogether, deriving pure unaccusatives, unmarked anticausatives, and raising verbs such as *scheinen* 'seem'.

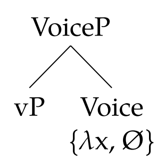
(3) **VoiceP and transitivity alternations in German**
(Adapted from Alexiadou et al. 2015, Schäfer 2008)

(a) Active thematic Voice



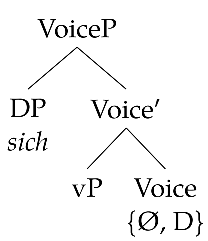
– Transitives and Unergatives

(b) Thematic non-active Voice



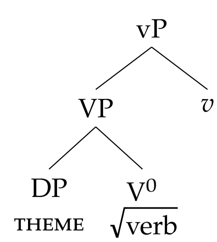
– Passives (*werden*)

(c) Expletive active Voice



– Marked anticausatives
– Reflexive middles

(d) Voiceless



– Pure unaccusatives
– Unmarked anticausatives
– Raising verbs (*scheinen*)

For purposes of definitional clarity: a **pure unaccusative** is a voiceless structure that never combines with an Agent, e.g. *erscheinen* 'appear'. An **an-**

ticausative is a change-of-state verb in a causative alternation; in their intransitive guise, these verbs appear i) in unaccusative form without voice layer as an unmarked anticausative, e.g. *zerbrechen* 'to break', or ii) as marked anticausatives with an expletive Voice layer merging reflexive *sich* in Spec,VoiceP (see Schäfer 2008). A **reflexive middle** is syntactically identical to the latter, making use of the same expletive active Voice layer but involving a different semantic intepretation. **Passive** structures employ a non-active thematic Voice. We now apply each diagnostic in turn.

2.2 *Basic detransitivization: passives and anticausatives*

In German, canonical passives are formed with the auxiliary *werden* and a past participle (Höhle 1978, Cysouw 2023); see (4). Passivization demotes the external argument while its Agent θ-role remains syntactically active (Baker, Johnson & Roberts 1989, Schäfer 2008, Alexiadou et al. 2015); the internal argument is promoted to subject. Because the Agent is still syntactically present, canonical passives license agent-oriented adverbs, permit control into purpose clauses, and allow the overt Agent in a *von*-phrase (Roberts 1987, Roeper 1987, Williams 1987, Bhatt & Pancheva 2006, Landau 2010, Bruening 2013, Alexiadou et al. 2015, Alexiadou, Anagnostopoulou & Schäfer 2018).

(4) a. *Der Wagen wird (von Anna) repariert.*
the.NOM car becomes (by Anna) repaired
'The car is repaired by Anna.'

The detransitivized counterpart of the passive is the anticausative. Anticausatives are change-of-state verbs in a causative alternation: they merge either as transitive causatives or as unaccusatives lacking an external argument. The unaccusative variant is structurally impoverished, lacking agentivity in both syntax and semantics (Schäfer 2008, Alexiadou et al. 2015), in contrast to the passive. One diagnostic for an anticausative is the felicity of the adjunct 'by itself' (Chierchia 2004), which is rendered as *von selbst* for German (Schäfer 2008: 56). This adjunct is barred from passives and middles. Moreover, under an adverb such as *leicht* 'easily', a non-agentive *at the slightest provocation* reading is available for anticausatives (Ackema & Schoorlemmer 2015, Pitteroff 2014). Conversely, *von*-phrases, agent-oriented adverbs, control into purpose clauses, and instrumental modifiers are all ungrammatical, since each requires a syntactically or semantically present Agent (Schäfer 2008, Alexiadou et al. 2015). *Für*-phrases are likewise unavailable in the relevant anticausative context, but their absence is not treated here as an independent Agent diagnostic (5).

(5) a. *Die Vase zerbrach (von selbst)...*
the vase broke (by itself)...
'The vase broke (by itself)...'

b. ...**von|*für Peter, *absichtlich, *mit dem Hammer, *um Peter eine Freude zu machen.*
...*by|*for Peter *deliberately *with the hammer *PURP Peter a joy to make.INF
' ...*by|*for Peter/ *deliberately/ *with the hammer/ *in order to make Peter happy.'

2.3 *The middle voice*

The middle voice sits between the passive and the anticausative. In Germanic, middles pattern syntactically with unaccusative anticausatives: they lack a syntactically active Agent (Hale & Keyser 1987, Roberts 1987, Fagan 1992, Ackema & Schoorlemmer 1995, 2015, Steinbach 2002, Lekakou 2005, Schäfer 2008, Pitteroff 2014) and therefore fail the agentivity tests a passive passes, rejecting *von*-phrases, agent-oriented adverbs, and control into purpose clauses. In German, canonical middles and reflexive-marked anticausatives are syntactically identical, both requiring *sich* (Schäfer 2008) (e.g. *sich öffnen* 'to open'), though semantically distinct. As with the anticausative, the middle also fails the *von selbst* test (Chierchia 2004, Schäfer 2008). These restrictions are shown in (6).[1]

(6) *Das Buch liest sich leicht (*von dem Mädchen) (*absichtlich) (*um etwas zu lernen) (*von selbst).*
the.NOM book.NOM reads REFL easily (*by the girl) (*intentionally) (*in.order something to learn) (*by [it]self)
'The book reads easily (*by the girl) (*intentionally) (*to learn something) (*by itself).'

German middles, however, do involve a semantically active Agent at the Conceptual-Intentional (C/I) interface (Lekakou 2005, Schäfer 2008, Pitteroff 2014). This conceptual Agent is most clearly visible in the agentive interpretation of facility adverbs and in instrumental adjuncts. *Für*-phrases can also occur in middles, and they are naturally related to the potential actor/ experiencer made available by the middle's semantics; what they do not show is that this participant is projected as a syntactic Agent. They have therefore been analysed either as related to the logical subject or as licensed by the ad-

1 Some speakers nonetheless accept modification by *von selbst* in middles.

verbial/modal semantics of the construction (Stroik 1992, Ackema & Schoorlemmer 1995, 2015) (7).

(7) *Das Buch liest sich gut (auf einem Kindle/ für Kinder).*
the book reads REFL well (on a Kindle/ for children)
'The book reads well (on a Kindle / for children).'

Taken together, (6) and (7) isolate the diagnostic signature of the middle. The first shows that no Agent is present in the narrow syntax: agentive *von*-phrases, agent-oriented adverbs, and control into purpose clauses all fail. The second shows that middle clauses nevertheless support material associated with a potential actor. Instrumental adjuncts and the *little effort* reading of facility adverbs provide the clearest evidence for a semantic Agent; *für*-phrases are compatible with that semantic profile as well, but they diagnose an entailed or modally accessible participant rather than a syntactically projected Agent (Fagan 1992, Ackema & Schoorlemmer 1995, 2015, Lekakou 2005, Pitteroff 2014). This split (syntactic Agent absent, semantic Agent present) is the property we use below to separate [*gehen*+*zu*-INFINITIVE] from both the passive and the anticausative.

Further evidence for the semantic Agent comes from examples ambiguous with reflexive-marked anticausatives, i.e. those that necessarily merge an expletive *sich* (Schäfer 2008). Here the Agent licenses a *little effort* reading (cf. Ackema & Schoorlemmer 2015, Pitteroff 2014) (8); the contrast between this and the anticausative's *slightest provocation* reading is the wedge separating a middle from an anticausative, and we return to it for [*gehen*+*zu*-INFINITIVE] in §2.5.

(8) *Die Tür öffnet sich leicht.*
the door opens REFL easily
Middle Reading: 'The door opens easily' (with little effort)
Anticausative Reading: 'The door opens easily' (at the slightest provocation)[2]

Finally, canonical middles show lexical-semantic constraints that bear directly on [*gehen*+*zu*-INFINITIVE]. They encode a modal dispositional reading, ascribing to the promoted internal argument a property that facilitates or impedes the event (Roberts 1987, Fagan 1992, Ackema & Schoorlemmer 1994, 1995, 2015, Steinbach 2002, Lekakou 2005), and they restrict the Vendlerian (1957) verb classes they admit: Germanic middles readily permit Accomplishments and Activities but are incompatible with Achievements and Statives (Roberts 1987, Fagan 1992, Ackema & Schoorlemmer 1994, 2015, Lekakou

2 A third, purely anticausative reading is available on which *leicht* is a degree modifier ('the door opens a little'), distinct from the 'at the slightest provocation' reading; we set it aside here.

2005). The middle's facility-adverb requirement (see Condoravdi 1989) demands that the promoted subject be a Patient or Theme (Ackema & Neeleman 2004, Ackema & Schoorlemmer 1995, 2015, Lekakou 2005) within a durative aspectual structure (cf. Fagan 1992), so that event progression can be mapped onto the subject (Condoravdi 1989). Accomplishments and Activities supply this durativity; Achievements, denoting an instantaneous change of state, and Statives, lacking dynamic progression, do not (see Lekakou 2005: 164–172). Statives additionally lack the agentive logical subject that middle formation relies on (Vendler 1957, Dowty 1979, Roberts 1987, Ackema & Schoorlemmer 1994), so they cannot entail the necessary Agent (cf. Condoravdi 1989, Schäfer 2008).

2.4 *Modal infinitives and restructuring in German*

Modal infinitives

German has a range of constructions that require a *zu*-infinitive. Structurally, [*gehen*+*zu*-INFINITIVE] resembles a well-documented family of infinitival detransitivization strategies known as "modal infinitives" (Brinkmann 1962) or "modal passives" (Höhle 1978): *sein* 'be' + [*zu*-INFINITIVE] (9a), and lexical verbs such as *bleiben* 'stay' (9b) or *stehen* 'stand' (see Haiden 2005, Holl 2010, Müller 2019, Abraham 2020). These form obligatorily coherent verb clusters that strongly resist extraposition (Haider 2010, Holl 2010).

(9) a. *Die Zugfenster sind nachts zu schließen.*
The train-windows are night-GEN to close-INF
'The train's windows must be closed at night.'

b. *Das Ergebnis bleibt abzuwarten.*
The result remains off-to-wait
'The result must be waited for.'

The literature debates where passivization is located in these structures. Three positions can be distinguished: (i) the marker *zu* is itself a passivizer that blocks Merge of an agentive external argument (for different mechanisms see Haiden 2005, Haider 2010, Holl 2010); (ii) *zu* is orthogonal to passivization, which instead follows from the infinitival configuration (see Demske-Neumann 1994) (a restructured, truncated VP (Wurmbrand 1998, 2001) or a specifierless VoiceP (Wurmbrand 2015)); or (iii) *zu* is not a passivizer at all, and passivization follows from the unaccusativity of the copula *sein* (Abraham 2020).

Sein+*zu* infinitives admit both a deontic and a dispositional (dynamic) possibility modality, i.e. *müssen* 'must' vs. *können* 'can' (Holl 2010). On the classic view, the deontic reading requires a syntactic Agent while dynamic possibility does not (Höhle 1978, von Stechow 1990, Bhatt 2006); the latter is often analysed as tough-movement yielding a middle reading (Steinbach 2002). The deontic version thus selects an agentive functional projection (e.g. VoiceP) hosting a covert Agent, whereas the possibility reading selects an agentless *v*P (Wurmbrand 2001).[3] This deontic/possibility split surfaces as a *von/für* asymmetry (10): the deontic reading licenses an agentive *von*-phrase, while the possibility reading favours a *für*-phrase. This asymmetry is the central distributional comparison we bring to [*gehen*+*zu*-INFINITIVE] in Section 3.

(10) The distribution of *von*- and *für*-phrases with *sein*+*zu*:

a. **Deontic:**

*Die Aufgaben sind (unbedingt) von allen/ *für alle zu lösen.*
the.NOM tasks.NOM are (absolutely) by all.DAT/ *for all.ACC ZU solve.INF

'The tasks (absolutely) must be solved by everyone/*for everyone.'

b. **Possibility:**

*Die Aufgaben sind (leicht) *von allen/für alle zu lösen.*
the.NOM tasks.NOM are (easily) *by all.DAT/for all.ACC ZU solve.INF

'The tasks can (easily) be solved *by everyone/for everyone.'
(ex.59 Holl 2010: 36, following Höhle 1978)

We therefore take the deontic modal passive to include an agentive VoiceP layer, while the possibility reading of *sein*+*zu* lacks one.

Restructuring

Modal infinitives like *sein*+*zu*, and [*gehen*+*zu*-INFINITIVE], must also be situated within Restructuring. A useful distinction separates Lexical Restructuring Infinitives (LRIs) from Functional Restructuring Infinitives (FRIs) (Wurmbrand 2001, 2004, Bobaljik & Wurmbrand 2005, Wurmbrand 2015). LRIs are governed by θ-assigning lexical verbs, e.g. *versuchen* 'try'; they involve control and may be active or passive. In (11), lexical restructuring permits long scrambling of the embedded object without cluster formation (the

3 Wurmbrand (2001) distinguishes *easy-to-please* from modal passives only by the presence of an adjective (lexical v_{sein} selecting AP vs. modal VAUX_{sein} selecting VP), and cannot derive the modal contrast itself; following Haider (2010: 309ff), we treat *leicht*, *schwer* 'easy, tough' in *sein*+*zu* as adverbials, with *sein* selecting a verbal complement.

'third construction', Wöllstein-Leisten 2001); A conservative assumption is a PRO in Spec,*v*P/VoiceP, with the matrix Agent controlling the embedded Agent. However, the mechanism of control is immaterial here; it has been modelled via PRO (Chomsky 1981, Landau 2013), post-syntactic Obligatory Control (Wurmbrand 2001), pre-syntactic argument pooling (Haider 2010), or Voice matching (Wurmbrand 2015).

(11)

weil	*er*	*den/*der*	*Traktor*	*versucht*	*hat*	[t_{OBJ}	*zu*	*reparieren*].
since	he	the.ACC/*the.NOM	tractor	tried	has		to	repair

'since he tried to repair the tractor'

(ex.5a Bobaljik & Wurmbrand 2005: 814)

FRIs, by contrast, involve a non-θ-assigning functional head in the matrix domain, e.g. *scheinen* 'seem'. These permit raising of the highest embedded argument but do not require control (12).

(12)

[*Der*	*Fehler*$_i$]	*scheint*	t_i	*zu*	*verschwinden.*
the.NOM	error.NOM	seems		ZU	disappear.INF

'The error seems to disappear.' **subj-to-subj raising**

The categorial status of functional restructuring heads such as *scheinen* is debated: high functional operators in the clausal spine (Cinque 1999, 2006a, Wurmbrand 2001, 2004, Roberts 2010), or unaccusative *v* heads selecting a reduced complement (Grewendorf 1989, Chomsky 2000, 2001, Davies & Dubinsky 2004, Sternefeld 2006, Haider 2010, Abraham 2020). Because both rely on selection of a defective domain, we model functional restructuring via an unaccusative *v* head; this avoids having to explain why a high auxiliary would idiosyncratically force a *zu*-infinitive rather than a bare infinitive, since the predicate's modal content can instead be encoded by an interpretable, valued modal feature on *v*, which we notate [i:Mod *val*] (an interpretable modality feature carrying a fixed value).

LRIs and FRIs also differ in extraposition. Raising *scheinen* (Haider 1993, Wurmbrand 2001) and *sein*+*zu* (Demske-Neumann 1994, Wurmbrand 2001, Haider 2010) both resist extraposition (13a,b), requiring strict coherence of the apparent cluster VRAISING+*zu*+VINFINITIVE. Control structures, and lexical restructuring verbs in long passives (the 'third construction'), do allow extraposition of the *zu*-infinitive away from the promoted subject (14) (Wöllstein-Leisten 2001, Müller 2020, Haider 2010).[4]

4 Wurmbrand (2001) is less accepting of this construction, but we follow the majority position.

(13) a. **dass Hans schien den Kuchen gegessen zu haben*
that John seemed the cake eaten to have.INF
'that John seemed to have eaten the cake'
(ex. 116n Wurmbrand 2001: 158)

b. **dass der Kuchen nicht ist zu essen*
that the.NOM cake.NOM not is zu eat.INF
'that the cake cannot/may not be eaten'
(ex. 1160 Wurmbrand 2001: 158)

(14) ... *dass [der Hund$_i$] vergessen wurde, [t_i zu füttern].*
... that the dog.NOM forgotten was to feed.
'that it was forgotten to feed the dog.' (ex.2c Haider 2010: 285)

These diagnostics (compound tense, extraposition, embedded auxiliaries, the *von/für* asymmetry, datives and double objects, adverbial modification, and Vendlerian class) give us the metrics needed to fix the status of [*gehen*+*zu*-INFINITIVE].

2.5 *The theoretical intersection of gehen*

With the baselines in place, we can locate [*gehen*+*zu*-INFINITIVE] among them. On the surface, the construction looks like a modal *zu*-infinitival passive (15a); semantically, it carries the strong dispositional, dynamic (im)possibility modality characteristic of middles and tough-movement. Yet, unlike the canonical middle of §2.3, it lacks the reflexive *sich* that German middles require (15b).

(15) a. *Aber jetzt geht es nicht mehr zu ändern.*
but now goes it not more to change
'But now it cannot be changed anymore.'

b. *Der Tisch geht leicht (*sich) abzuwischen.*
the.NOM table goes easy (*REFL) off-to-wipe
'The table wipes easily / can be easily wiped clean.'

The most detailed description to date, Cysouw (2023), labels the construction an *Unmöglichkeitsantikausativ* 'impossibility anticausative', capturing two observations: (i) an apparent loss of causative behaviour, i.e. promotion of the Theme to subject; and (ii) the frequent presence of negation, invoking a dispositional property of the subject that impedes the action (16a). However, the construction is neither limited to impossibility (16b,c) nor classifiable as an anticausative.

(16) a. *Dieses Glas geht nicht zu schneiden, weil es Sicherheitsglas ist.*
this glass goes not to cut because it safety-glass is
'This glass cannot be cut because it's safety glass'

b. *Die temporäre Datei geht zu löschen, aber nicht die exe.*
the temporary file goes to delete but not the exe
'The temporary file can be deleted but not the exe.'

c. *Das Radio geht zu reparieren.*
the radio goes to repair
'The radio can be repaired' (see examples in Cysouw 2023)

Applying the anticausative diagnostic of §2.2 confirms that the label is wrong. A true anticausative, lacking any Agent, freely admits *von selbst* (cf. Chierchia 2004); but [*gehen*+*zu*-INFINITIVE] categorically rejects it (17). Moreover, [*gehen*+*zu*-INFINITIVE] elicits the middle's *with little effort* reading and disallows the anticausative's *at the slightest provocation* reading (see Pitteroff 2014, Ackema & Schoorlemmer 2015). Both diagnostics align [*gehen*+*zu*-INFINITIVE] with the middle and show that an Agent is at least semantically active; the construction is therefore not a periphrastic anticausative (contra Cysouw 2023).

(17) a. *Die Tür geht leicht (*von selbst) zu öffnen.*
the.NOM door goes easy (*by [it]self) to open.INF
'The door can be opened easily (*by itself)'

Cysouw (2023) also offers a short list of attested verbs from a corpus search (18).

(18) **Attested verbs**
ändern 'to change', *bauen* 'to build', *beheben* 'to rectify', *löschen* 'to extinguish', *reparieren* 'to repair', *rezipieren* 'to reciprocate', *schneiden* 'to cut', *stopfen* 'to stuff', *verschließen* 'to lock' (Cysouw 2023)

The 12 attested examples are too few to settle the construction's behaviour, but a testable tendency emerges: the verbs in (18) appear restricted to Accomplishments (Vendler 1957). Coupled with the dispositional property ascribed to the Theme, this aligns [*gehen*+*zu*-INFINITIVE] most closely with established medio-passive constructions (canonical middles and *lassen*-middles) and yields a clear hypothesis: [*gehen*+*zu*-INFINITIVE] is a non-reflexive, periphrastic middle. Consistent with this, our own and consultant judgements reject canonical Achievements (19a) and Statives (19b) under *ge-*

hen, both of which are independently barred from middles.

(19) a. **Der Autoschlüssel geht nicht/leicht zu finden.*
the car-keys go not/easy to find

b. **Die Welt geht nicht zu kennen.*
the world goes not to know

By delineating the syntactic and aspectual boundaries of passives, anticausatives, and middles, we have assembled a set of diagnostics that can now be applied to [*gehen*+*zu*-INFINITIVE] in depth. The baselines yield two competing hypotheses with sharply different predictions. If [*gehen*+*zu*-INFINITIVE] is a true periphrastic passive on the model of [*sein*+*zu*-INFINITIVE], it should license a covert but syntactically active Agent and the phenomena that depend on one: agentive *von*-phrases, agent-oriented adverbs, and control into purpose clauses. If, instead, [*gehen*+*zu*-INFINITIVE] realizes the middle voice, the syntactic Agent should be absent while the construction remains dispositionally oriented: it should require disposition-aligned modification (*easily*, *simply*, *tough*, *not*), reject Vendlerian Achievements, datives, and ditransitives, permit instrumental modifiers, and favour *für*-phrases over agentive *von*-phrases. The next section subjects [*gehen*+*zu*-INFINITIVE] to a quantitative acceptability study built around exactly these parameters.

3 Experiment

Section 2.5 left us with two competing hypotheses: [*gehen*+*zu*-INFINITIVE] as a periphrastic passive versus [*gehen*+*zu*-INFINITIVE] as a periphrastic realization of the middle voice. To decide between them, we ran a quantitative acceptability judgment *pilot* study for [*gehen*+*zu*-INFINITIVE], targeting the core diagnostic properties that distinguish raising, passive, and middle structures. We recruited 72 L1-German participants to rate 50 randomized stimuli (presented with context) on a 1–5 Likert scale. The stimuli tested various conditions placed on [*gehen*+*zu*-INFINITIVE] itself and compared it to other semantically or syntactically aligned structures, including *lassen*-middles, canonical *werden*-passives, and modal *zu*-infinitives with *sein*.

The experimental design specifically targeted the core diagnostic properties capable of differentiating raising, passive, and middle structures. These included structural restrictions at the TP and *v*P levels (periphrastic tense, extraposition, and dative/double object preservation) (Haider 1993, 2010, Wurmbrand 1999, 2001), syntactic agentivity and modal/dispositional properties (*von*- vs. *für*-phrases, instrumental adjuncts) (Keyser & Roeper 1984, Roberts 1987, Baker et al. 1989, Fagan 1992, Ackema & Schoorlemmer 1995, 2015), and lexical-semantic aspectual restrictions (cf. Vendler 1957).

Linear Mixed-Effects Models were fitted with crossed random intercepts for participants and items.[5] Fixed effects were sum-coded for factorial designs; treatment coding was used for multi-level single factors, with canonical constructions as the reference level. Descriptive statistics for all conditions are visualized in the accompanying figures. In prose, we report standardized means (M_z) alongside inferential model output (β, SE, z, p).

3.1 *Establishing relative acceptability thresholds*

Non-standard constructions systematically suffer from prescriptive bias in formal tasks, resulting in deflated ratings (Schütze 1996, Cornips & Poletto 2005). Two structural anchors were established for interpreting gradient well-formedness (cf. Featherston 2005, 2007). The absolute experimental ceiling ($M = 4.63, M_z = 1.19$) was defined via grammatical canonical passive fillers ($n = 9$), while the floor ($M = 1.32, M_z = -1.04$) used ungrammatical controls ($n = 3$). The highest-rated *gehen* item ($M = 3.85, M_z = 0.64$) serves as the construction-specific ceiling (C_{ns}), revealing a prescriptive penalty of approximately 0.78 raw scale points ($\Delta_z = 0.55$) (Table 1).

Tier	**Canonical Ceiling** $M_z = 1.19$	**Non-standard Ceiling** $M_z = 0.64$
Fully Licensed	$M_z \geq 0.65$	$M_z \geq 0.40$
Marginal	$-0.15 \leq M_z < 0.65$	$-0.30 \leq M_z < 0.40$
Rejection	$M_z < -0.15$	$M_z < -0.30$

Note. Raw-scale equivalents: canonical fully licensed $M \geq 3.90$, marginal $2.60 \leq M < 3.90$, rejection $M < 2.60$; non-standard fully licensed $M \geq 3.50$, marginal $2.50 \leq M < 3.50$, rejection $M < 2.50$.

Table 1 Relative Acceptability Thresholds by Construction Type

In all boxplots, dashed horizontal lines indicate standardized thresholds of acceptability: The green dashed line marks the canonical ceiling ($M_z = 1.25$); the red dashed line indicates the experimental floor ($M_z = -1.04$). The blue dotted line indicates the threshold for fully licensed *gehen* items ($M_z = 0.40$), and the purple dash-dotted line marks its theoretical rejection threshold ($M_z = -0.30$). Appendix A provides tables with the raw data for each test; the complete list of all 50 experimental stimuli, with their raw and *z*-score standardized acceptability means, is given in Appendix B.

5 Statistical modeling and data visualization were performed in Python 3 using the `statsmodels` library.

3.2 Experimental Data: Results

3.2.1 Structural restrictions: tense, extraposition, and arguments

Compound tense

We tested the acceptability of synthetic and periphrastic morphology for *gehen* compared to the canonical raising predicate *scheinen*, which resists compound tense morphology (Haider 1993, 2010, Wurmbrand 2001). The contrasts are given in (20–21).

(20) a. *Die Journalistin schien gut informiert zu sein.*
The journalist seemed well informed to beINF
'The journalist seemed to be well informed.'

b. **Die Journalistin hat gut informiert zu sein geschienen.*
The journalist has well informed to be seemed.PTCP

(21) a. *Der Wagen ging nicht zu reparieren.*
The car went not to repair.INF
'The car could not be repaired.'

b. **Der Wagen ist nicht zu reparieren gegangen.*
The car is not to repair gone.PTCP

Figure 1 shows that simple forms of *scheinen* ($M_z = 1.39$) and *gehen* ($M_z = 0.42$) are in the licensed ranges, yet periphrastic forms are ungrammatical; both *scheinen* ($M_z = -0.94$) and *gehen* ($M_z = -0.95$) cluster at the experimental floor ($M_z = -1.04$).

An LMM revealed significant main effects for MORPHOLOGY ($\beta = 2.76$, $SE = 0.10$, $z = 26.41$, $p < .001$) and VERB ($\beta = 0.72$, $SE = 0.10$, $z = 6.85$, $p < .001$), alongside a significant interaction ($\beta = 1.38$, $SE = 0.21$, $z = 6.59$, $p < .001$). The prescriptive penalty against *gehen* is restricted entirely to simple tenses. In compound tenses, both verbs categorically collapse to the experimental floor and are statistically indistinguishable ($\beta = 0.03$, $SE = 0.12$, $z = 0.23$, $p = .815$). This confirms that *gehen* patterns with a functional raising head like *scheinen* (cf. Wurmbrand 2001).

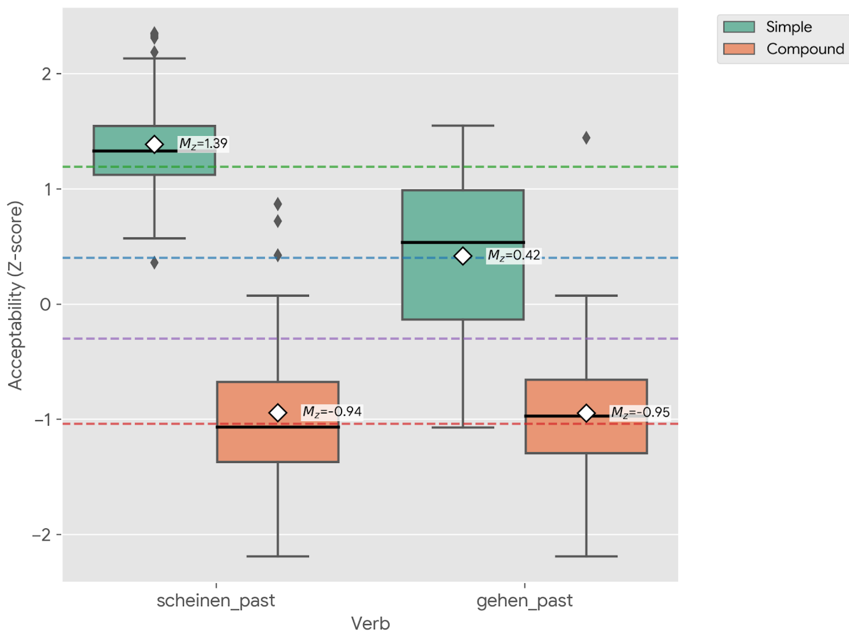


Figure 1 Distribution of acceptability (Z-scores) for simple and compound tense forms

Extraposition

The extraposition test distinguishes lexical heads, which permit rightward displacement of the *zu*-infinitive (Wöllstein-Leisten 2001, Müller 2020, Haider 2010), from functional restructuring heads, which require a coherent intraposed cluster (Haider 1993, 2010, Wurmbrand 2001). Extraposition of [*zu*-INFINITIVE] is definitively judged as ungrammatical ($M_z = -1.03$), falling at the experimental floor ($M_z = -1.04$), while the intraposed form ($M_z = 0.30$) falls close to the fully licensed range (22); see Figure 2.

(22) *Louisa ist sehr glücklich ...*
('Louisa is very happy ...')
dass das Spiel endlich *(*zu installieren*) *geht* (**zu installieren*).
that the game finally (to install) goes to install
'that the game finally can be installed.'

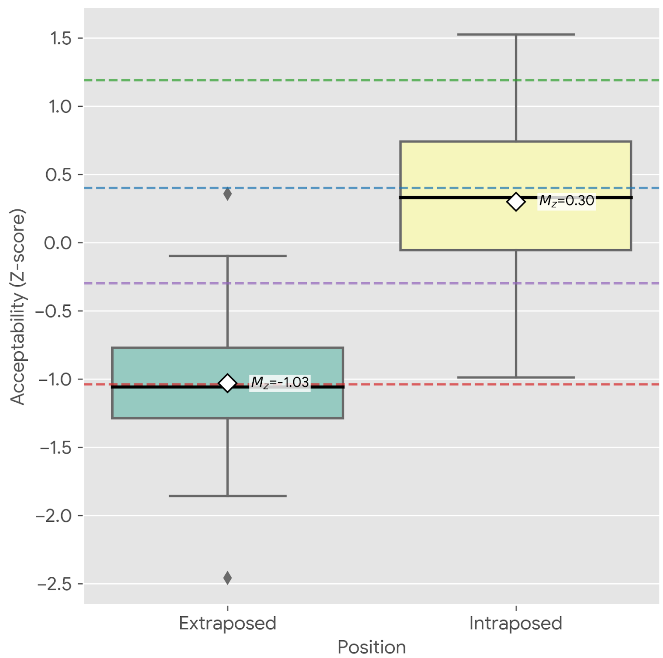


Figure 2 Distribution of standardized acceptability (Z-scores) for intraposed vs. extraposed infinitives with *gehen*

An LMM using treatment coding (with the intraposed condition as the reference level) confirmed this collapse in acceptability to be highly significant ($\beta = -2.00$, $SE = 0.14$, $z = -14.30$, $p < .001$). *Gehen* thus behaves as a functional raising head requiring a coherent infinitive structure, patterning with the functional raising verb *scheinen* and modal passive [*sein*+*zu*-INFINITIVE], and diverging from lexical restructuring heads in long passives, which freely permit rightward displacement of the infinitival complement.

Datives and double arguments

Dative and ditransitive predicates provide a further test of the size and internal structure of the infinitival complement, though not a simple middle/passive diagnostic. Modal passives and tough-movement constructions permit dative preservation (Fagan 1992, Ackema & Schoorlemmer 1995, Steinbach 2002), and German middles can preserve datives in special configurations, including impersonal *es*-middles (Steinbach 2002). We therefore compared dative-assigning *helfen* ('help') across *sein* and *gehen* configurations to test whether [*gehen*+*zu*-INFINITIVE] permits the relevant dative structure (23);

the *sein* form here is a modal *sein+zu* passive, not a tough-construction.

(23) *Ihm* *ist/*geht* *nicht* *zu* *helfen.*
him.DAT is/goes not to help.INF
'He cannot/should not be helped.'

As visualized in Figure 3, the *sein*-passive remains around the canonical ceiling ($M_z = 1.32$), while *gehen* is categorically rejected, clustering at the absolute experimental floor ($M_z = -0.99$). An LMM using treatment coding (with [*sein+zu*-INFINITIVE] as the reference level) confirmed a total collapse in acceptability ($\beta = -3.43$, $SE = 0.11$, $z = -31.11$, $p < .001$).

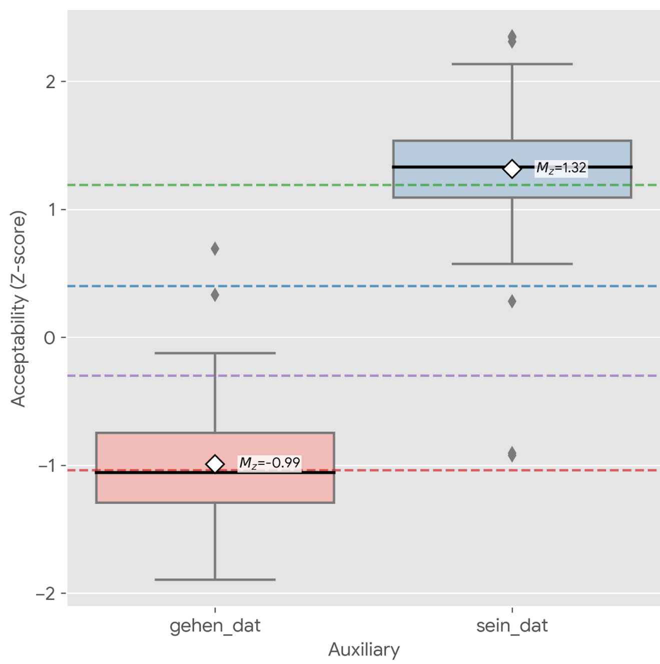


Figure 3 Acceptability (Z-scores) of *sein* vs. *gehen* with a dative-assigning verb *helfen*

However, there is a methodological confound in the *gehen* condition regarding the monotransitive stimulus *helfen* ('to help'). As established in Section 2, *gehen* prefers Accomplishment predicates. Since *helfen* is an Activity, it was likely subjected to a double penalization: a violation of an aspectual requirement and a potential structural violation. This confound cannot be resolved by substituting the verb, since German systematically lacks dative-assigning two-place Accomplishments. Dative resultatives appear limited to punctual Achievements (e.g., *begegnen* 'to meet', *entkommen* 'to escape', *auffal-*

len 'to stand out'), which are independently incompatible with middles and *gehen*, as we will show. Consequently, the monotransitive data alone cannot prove a "dative allergy".

The ditransitive condition is therefore more informative, not because ditransitives are independently impossible in all German middles, but because it bypasses the aspectual confound: *schicken* 'send' is a ditransitive Accomplishment. While ditransitive predicates are grammatical with the modal [*sein*+*zu*-INFINITIVE]-passive (24), *gehen* rejects them (25).[6] Indeed, the ditransitive *gehen* stimulus yielded one of the lowest ratings ($M_z = -1.14$).

(24) *Das Buch ist ihm/dem Eigentümer zurückzugeben/auszuhändigen.*
the book is him/the.DAT owner back-to-give/hand-to-over
'The book must be given/handed over to the owner.'

(25) **Die Email geht ihm nicht zu schicken.*
the email goes him not to send
Intended: 'The email cannot be sent to him.'

3.2.2 *Semantic and syntactic agentive properties*

Passives are well known to pass diagnostics for a syntactically active agent, while canonical middles in Germanic pass only tests for a semantically active Agent (Roberts 1987, Baker et al. 1989, Fagan 1992, Steinbach 2002, Lekakou 2005, Ackema & Schoorlemmer 1995, 2015, Pitteroff 2014, Alexiadou et al. 2015).

Agentive von-phrases

Canonical passives (including modal *sein*+*zu*, see 26) unproblematically license agentive *von*-phrases, signalling a syntactically active but demoted external argument (Baker et al. 1989), yet canonical middles categorically reject them (Ackema & Schoorlemmer 2015). We tested the compatibility of *gehen* with a *von*-phrase against these established baselines to diagnose the syntactic status of its implicit Agent.

6 The *sein* and *gehen* ditransitive items differ in verb and in polarity, but this is incidental: what is at stake is the (un)availability of the relevant applicative/second-internal-argument configuration under *gehen*, irrespective of verb choice or negation.

(26) *Das* *Ergebnis* *eines* *demokratischen* *Prozesses* *sei*
the.NOM result.NOM a.GEN democratic.GEN process.GEN be.SBJV.3SG
von *allen* *zu* *respektieren.*
by all.DAT to respect.INF
'The result of a democratic process is to be respected by everyone.'
(DWDS, *Berliner Zeitung* 02.12.2004)

gehen with a *von*-phrase (27) is categorically rejected ($M_z = -0.67$), representing a massive structural penalty ($\Delta M_z = 1.31$) compared to baseline *gehen*-middles. The data are visualized in Figure 4.

(27) **Die* *Tür* *geht* *von* *Anna* *kaum* *aufzuschließen.*
the.NOM door.NOM goes by Anna barely up-to-lock
'The door can barely be unlocked by Anna.'

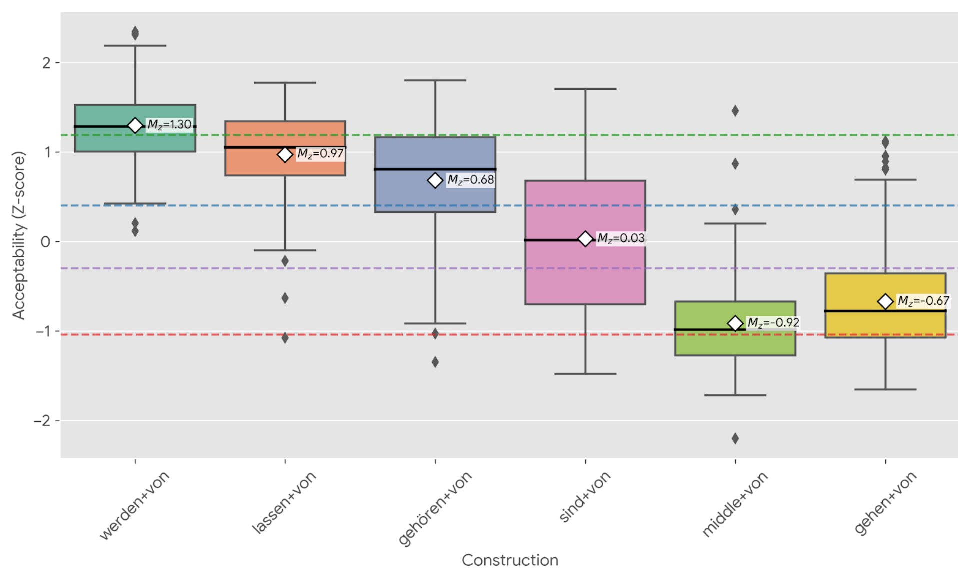


Figure 4 Distribution of acceptability (*Z*-scores) for agentive *von*-phrases across passive and middle constructions.

An LMM using treatment coding (with *gehen* set as the reference baseline) confirmed highly significant structural divergence from all passive structures. Canonical *werden*-passives ($\beta = 2.88$, $SE = 0.10$, $z = 30.25$, $p < .001$), *lassen*-passives ($\beta = 2.47$, $SE = 0.12$, $z = 20.51$, $p < .001$), *gehören*-passives ($\beta = 2.04$, $SE = 0.12$, $z = 16.93$, $p < .001$), and modal *sein+zu* passives ($\beta = 1.02$, $SE = 0.12$, $z = 8.50$, $p < .001$) all license explicit agents with significantly higher acceptability. Crucially, *gehen* patterns strictly with the agentless reflexive middle, which exhibits a similarly deep rejection ($\beta = -0.38$, $SE = 0.12$, $z = -3.15$, $p = .002$).

Für-phrases and middle-like modality

While canonical middles categorically reject agentive *von*-phrases, they often license *für*-phrases. We take this contrast to be agent-related, but not in the same way as a *von*-phrase: *für*-phrases are compatible with an entailed or modally accessible participant, but they do not diagnose a syntactically projected Agent. In the middle literature, *for*-phrases have accordingly been analysed either as expressions of the logical subject or as experiencer/benefactive phrases licensed by adverbial/modal semantics (Stroik 1992, Ackema & Schoorlemmer 1995, 2015). We therefore compared *für*- vs. *von*-phrase compatibility across *gehen*, canonical reflexive middles, and modal *sein+zu* passives. The *für*-variant of the rejected *von*-phrase above is comparatively acceptable (28).

(28) *Die Tür geht für Anna kaum aufzuschließen.*
the.NOM door.NOM goes for Anna barely up-to-lock
'The door can barely be unlocked for Anna.'

As shown in Figure 5, a *für*-phrase raises *gehen* from categorical rejection with a *von*-phrase ($M_z = -0.67$) into the marginally acceptable zone ($M_z = -0.08$). Similar but stronger contrasts were observed in the canonical constructions. Canonical middles heavily preferred *für*-phrases ($M_z = 0.54$) over *von*-phrases ($M_z = -0.92$), and *sein+zu* structures operated at the canonical ceiling with *für*-phrases ($M_z = 1.23$) while degrading to marginality with *von*-phrases ($M_z = 0.03$). The *sein+zu* item may itself favour the *für*-variant: [*sein+zu*-INFINITIVE] allows both a highly agentive deontic reading and a possibility interpretation (Höhle 1978, Steinbach 2002, Holl 2010), while *sehen* ('see') favours the latter without categorically excluding deonticity.

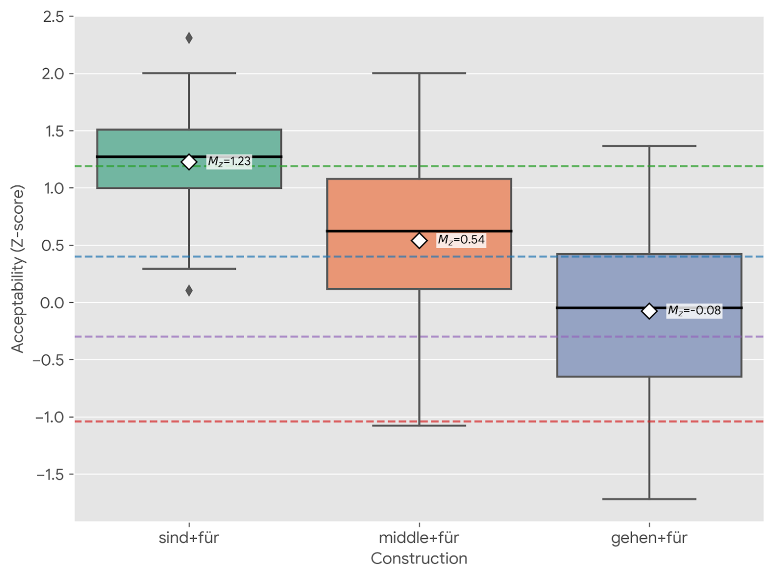


Figure 5 Distribution of acceptability (Z-scores) across three target constructions comparing explicit agentive *von*-phrases with *für*-phrases

Regardless, an LMM using treatment coding (setting *gehen* and *von* as reference baselines) tested the interaction between VERB CONSTRUCTION and ADJUNCT TYPE. The model confirmed a highly significant main effect of ADJUNCT: across all constructions, *für*-phrases are significantly preferred over explicit *von*-phrases ($\beta = 0.89$, $SE = 0.09$, $z = 9.64$, $p < .001$).

The "acceptability gap" between *für* and *von* is significantly narrower for *gehen* than for both the canonical middle ($\beta = 1.29$, $SE = 0.19$, $z = 6.94$, $p < .001$) and the *sein* passive ($\beta = 0.93$, $SE = 0.19$, $z = 5.00$, $p < .001$). We attribute this to speaker insecurity stemming from a form-meaning mismatch: *gehen* resembles modal *zu*-infinitives, which slightly softens the penalty for *von*-phrases compared to the unambiguous reflexive middle. Distributionally, however, the preference for *für* over *von* aligns *gehen* with middle and tough-type possibility readings rather than with passives that freely introduce explicit agents.

Instrumental adjuncts

The semantic presence of a syntactically inactive implicit Agent in middles generally licenses instrumental adjuncts (Pitteroff 2014). To verify whether *gehen* aligns with this property, we tested a single stimulus containing an instrumental phrase without an explicit Agent (29).

(29) *Die Schraube geht aber mit einem besonderen Schraubenzieher*
the.NOM screw.NOM goes but with a.DAT special.DAT screwdriver
leicht zu lösen.
easily to loosen
'The screw can easily be loosened with a special screwdriver.'

An instrumental adjunct with [*gehen*+*zu*-INFINITIVE] fell securely within the marginally licensed zone for *gehen* ($M_z = 0.21$). Because only a single item was tested, an intercept-only regression model was used to compare the raw ratings against the absolute experimental floor ($M = 1.32$). The model confirmed that the instrumental condition is structurally licensed significantly above the categorical rejection threshold ($\beta = 1.89$, $SE = 0.16$, $t = 11.85$, $p < .001$).

The felicity of the instrumental adjunct indicates that [*gehen*+*zu*-INFINITIVE] licenses an implicit Agent at the semantic interface, consistent with a periphrastic middle analysis. On this account, agent-oriented adverbs should pattern with *von*-phrases in being ungrammatical, a prediction we address qualitatively in Section 4.

3.3 Lexical-semantic restrictions

Adverbial Modification

If *gehen* functions as a middle, it should exhibit preferences for facilitative adverbial modification (Roberts 1987, Condoravdi 1989, Schäfer 2008). To map its licensing profile, we tested a range of adverbial and negative modifiers. The results, visualized in Figure 6 (see Table 9 in Appendix A for full descriptive statistics), confirm that *gehen* shows a very strong alignment with scalar modification, an alignment we argue has tightened through grammaticalization, setting it apart from canonical passive structures.

The hierarchy indicates that *gehen* is primarily licensed by modal and degree qualification. Modifiers explicitly linked to scalar potential (*kaum*, *leicht*, *einfach*) elevated the construction securely into the fully licensed zone. Similarly, the relative success of restitutive *wieder* (30) stems from its function of targeting a predicate's result state (Von Stechow 1996, Dobler 2008, Alexiadou & Schäfer 2011); The subject's inherent dispositional potential as a gradable property is facilitated by presupposing the restoration of a previously held state.[7]

7 Scores for *wieder* revealed a bimodal distribution, suggesting a divergence between speakers who successfully accessed a restitutive reading and likely those who interpreted it as temporal

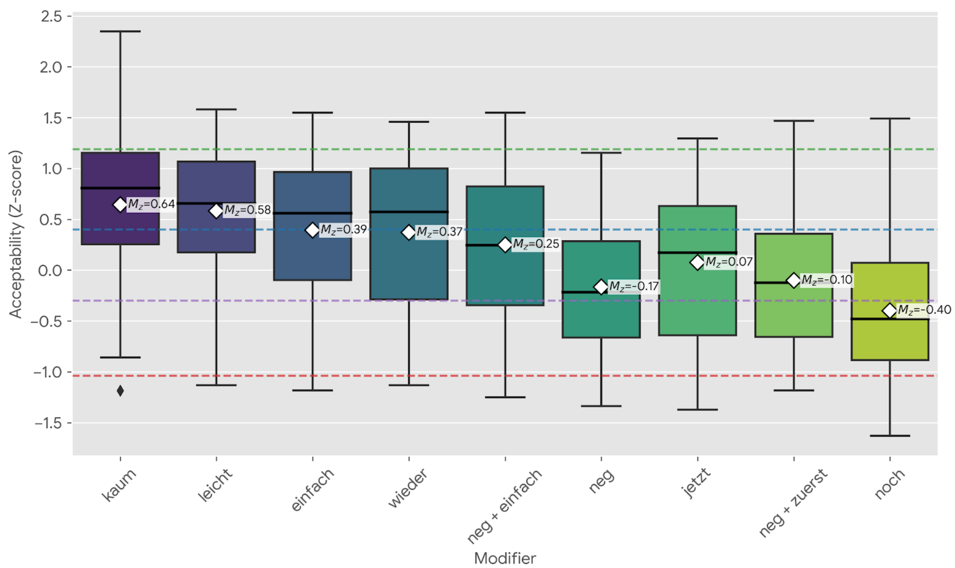


Figure 6 Mean acceptability *Z*-scores for adverbial/negative modification of *gehen*

(30) *Die* *Tür* *geht* *jetzt* *wieder* *auf-zu-schließen.*
the.NOM.SG.F door go.3SG.PRES now again up-to-close.INF
'The door can be locked/unlocked again now.'

This modal licensing requirement disproves that [*gehen*+*zu*-INFINITIVE] is fundamentally associated with negation as an "impossibility anticausative " (Cysouw 2023), in addition to showing general behaviour inconsistent with an anticausative analysis. The experimental results demonstrate that bare negation (*gehen_neg*) provides only a marginal improvement ($M_z = 0.14$). In fact, adding a modal adverb to a negated structure (*einfach nicht*, $M_z = 0.25$) measurably boosts acceptability over negation alone. Negation cannot by itself "rescue" an otherwise illicit structure; it helps only insofar as it forces a modal reading, a behaviour consistent with canonical middles (for middles see Roberts 1987, Condoravdi 1989, Schäfer 2008, Lekakou 2005). The modal and scalar potential structurally anchors the construction.

Conversely, purely temporal framing proved insufficient. The adverbs *jetzt* and *zuerst* landed firmly in the marginal/rejected zones, demonstrating widespread structural failure. The clearest degradation occurred with the modifier *noch* 'still', which we judge highly marked rather than fully ungram-

repetitive marker.

matical (31).

(31) *??Der Quark geht noch zu essen.*
the.NOM quark.NOM go.PRES.3SG still to eat.INF
'The quark is still edible / can still be eaten.'

This degradation sheds light on *gehen*'s lexical-semantic constraints. Although verbs like *essen* 'eat' can be compositionally coerced into telic Accomplishments (Vendler 1957), the root itself is an atelic manner-activity verb (see MANNER/RESULT COMPLEMENTARITY in Rappaport Hovav & Levin 2010). Unlike evaluative and degree modifiers, *noch* is too weak to license a scale independently: it presupposes a pre-existing scalar or phasal structure supplied by the verb (König 1991: 52–56). With a manner root like *essen*, this presupposition fails ($M_z = -0.40$). It should succeed, however, with inherently resultative roots such as *reparieren* 'repair', as in the natural example in (32).[8] This confirms that *gehen* interacts with the lexical aspect of its complement, a restriction we unpack below.

(32) *vieles geht noch zu reparieren!*
much.NOM goes still to repair.INF
'Many things can still be repaired!'

To statistically evaluate this hierarchy, an LMM with a planned contrast was used (setting temporal/aspectual modifiers as the reference baseline). The model confirmed that modal and degree modifiers generated a massive, highly significant structural advantage ($\beta = 1.04$, $SE = 0.09$, $z = 11.20$, $p < .001$). While canonical middles are generally restricted to evaluative manner adverbs, the unique preference of *gehen* for degree adverbs (*kaum*) and restitutive aspect (*wieder*) empirically confirms its specialization as a functional raising head measuring the realisability of a potential state.

Vendler's verb classes

Canonical middles show a preference for Accomplishments, are compatible with Activities, but categorically reject Achievements (Roberts 1987, Fagan 1992, Ackema & Schoorlemmer 1995, Lekakou 2005, Pitteroff 2014). To test whether *gehen* observes these same aspectual restrictions, we analysed acceptability scores across all stimuli representing Accomplishments ($n = 6$), Activities ($n = 3$), and Achievements ($n = 5$). The results confirmed a strong preference for inherent Accomplishments (33a), while Achievements were categorically rejected (33b) (see also Figure 7).

8 Source: Nissanboard, https://www.nissanboard.de/forum/index.php?thread/100011-problem-mit-der-hinterachse-an-meinem-gti-hilfe/&postID=2033330. Posted October 24, 2008.

(33) a. *Die Tür geht jetzt leicht aufzuschließen.*
The door goes now easy up-to-close.INF
'The door can easily be unlocked now.'

b. **Der Schaffner geht nicht zu überzeugen.*
the conductor go not to convince.INF
Intended: 'The conductor cannot be convinced.'

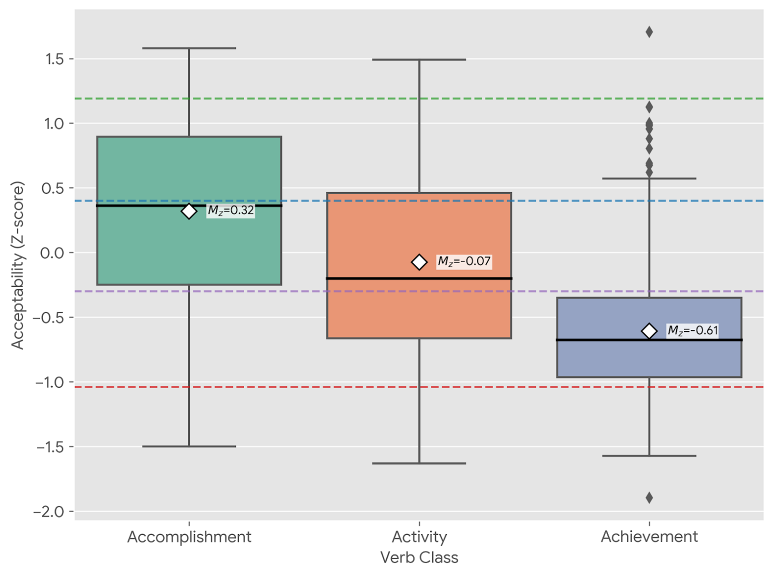


Figure 7 Distribution of acceptability (Z-scores) across verb classes under *gehen*

Because scalar adverbial boosters (e.g., *leicht*, *einfach*) were unevenly distributed across the verb classes in our stimuli, raw averages risk conflating the aspectual capacity of the verb with the independent licensing power of the adverb. An LMM incorporating adverbial type as a fixed-effect covariate was therefore used to partial out this variance and isolate the underlying aspectual pattern.

Even after controlling for the independent contribution of adverbial modification ($\beta = 0.47$, $SE = 0.08$, $z = 5.54$, $p < .001$), the aspectual hierarchy remained intact. Accomplishments ($M_z = 0.32$) significantly outperformed Activities ($M_z = -0.07$; $\beta = 0.45$, $SE = 0.09$, $z = 4.81$, $p < .001$). Both Accomplishments ($\beta = 1.25$, $SE = 0.08$, $z = 16.59$, $p < .001$) and Activities ($\beta = 0.80$, $SE = 0.11$, $z = 7.48$, $p < .001$) showed substantially higher acceptability than Achievements ($M_z = -0.61$).

The aspectual preference persists under adverbial boosting: *spielen* 'to play' reaches the licensed zone with *leicht* ($M_z = 0.34$, 34a), but still underperforms an equivalently boosted Accomplishment such as *aufzuschließen* 'to unlock' ($M_z = 0.58$, 33b). Future work should test a wider range of manner roots.

(34) *Jimmys Gitarre geht leicht zu spielen.*
Jimmy's guitar goes easy to play.INF
Intended: 'Jimmy's guitar plays easily.'

Achievements resist adverbial boosting entirely. Even when intensified by the evaluative scalar adverb *einfach* ($M_z = -0.69$), they remain overwhelmingly less acceptable than standard, non-intensified Accomplishments ($M_z = 0.14$). This confirms that the lexical-semantic constraint against punctual (and thus ungradable) Achievements overrides the licensing power of adverbs.

Together, these results indicate that *gehen* requires a complement encoding a gradable result state. The punctual structure of Achievements provides no scalar domain for the dispositional operator to evaluate, accounting for their categorical rejection.

4 Discussion

We now address the theoretical implications of the results, organized by domain: clausal syntax, argument structure, and lexical semantics. We argue that *gehen* is a functional raising head that enforces detransitivization through unaccusative syntax. Since the experiment was designed as a pilot study, we supplement the quantitative results where necessary with post-hoc native-speaker judgments from related (medio-)passive configurations.

4.1 Clausal Architecture and Selection

Evidence for the functional status of *gehen* stems from its resistance to extraposition. This aligns it with functional raising verbs and modal passives, which form a coherent verb cluster or single prosodic phrase with their complement (see Haider 2010, Wurmbrand 2001: 335f), and distinguishes it from LRIs in Long Passives, which can permit extraposition (see also Wöllstein-Leisten 2001, Haider 2010, Müller 2020). We understand functional status here as the inability to assign thematic roles, not as a requirement that *gehen* lexicalize a non-verbal functional projection in T.

(35) *Louisa is very happy that ...*
...**dass [das Spiel] endlich geht [zu installieren].*
...that [the game.NOM] finally [goes] [to install].
'...that the game can finally be installed.'

(36) *... dass [der Hund$_i$] vergessen wurde, [t_i zu füttern].*
... that the dog.NOM forgotten was to feed.
'that it was forgotten to feed the dog.' (ex.2c Haider 2010: 285)

The size of this complement is further constrained by auxiliary selection: auxiliaries in the infinitival domain of *gehen* are strongly rejected (37).

(37) **Das Fahrrad geht (bis morgen) leicht repariert zu sein/haben.*
The bike goes (by tomorrow) easily repaired to be/have.INF
Intended 'The bike can be easily repaired by tomorrow.'

We note that *scheinen* allows auxiliaries after *zu* (38a), while modal *sein+zu* (Haiden 2005, Holl 2010) (38b) and LRIs are incompatible with auxiliaries in the infinitival domain (38c).

(38) a. *Sie scheint das Buch gelesen zu haben.*
She seems the book read.PST.PTCP to have.AUX.INF
'She seems to have read the book.'

b. **Sie ist (bis morgen) fotografiert zu sein / zu haben / worden zu sein.*
she is (by tomorrow) photographed to be / to have / been to be
Intended: 'She must have been to be photographed by tomorrow.'
(ex.30 Holl 2010: 49)

c. ...**dass schon [der Traktor und der Lastwagen] repariert zu haben vergessen wurden.*
...that already [the tractor and the truck.NOM] repaired to have.INF forgotten were.PASS.3SG.PL
'that it was forgotten to have already repaired the tractor and the truck.' (ex.240b Wurmbrand 2001: 300)

The difference follows if *scheinen* selects TP, while LRIs and *sein+zu* take a truncated, voiceless *v*P complement (see Wurmbrand 2001: 3, 79). The auxiliary ban therefore indicates that *gehen* also selects a *v*P without higher functional structure (see Wurmbrand 2001, Holl 2010, Haider 2010).

Gehen also restricts its upper domain: matrix compound tense morphology is ungrammatical, as with *scheinen*. For *scheinen*, Wurmbrand (2001) derives this from its status as an epistemic modal above T. That cannot carry over directly: dispositional *gehen* is a dynamic possibility operator, and dynamic modals normally sit below T (Wurmbrand 2001, Cinque 2006b). Section 5 therefore derives the auxiliary ban through feature valuation rather than high modal scope.

Table 2 summarizes these clausal diagnostics across the four constructions: the modal passive *sein+zu*, *gehen+zu*, *scheinen+zu*, and LRIs in long passives.

Feature	***sein+zu***	***gehen+zu***	***scheinen+zu***	**LRI**
Compound Tense (Matrix)	✓	✗	✗	✓
Extraposition	✗	✗	✗	✓
Auxiliaries in inf. Domain	✗	✗	✓	✗

Table 2 A comparison of different restructuring constructions

4.2 *Lexical Semantics*

Lexical-semantically, [*gehen+zu*-INFINITIVE] is conditioned by two requirements: modal qualification by scalar potential and aspectual compatibility with a root/configuration encoding a result state.

Adverbial modification lets us reject an "impossibility anticausative" analysis (Cysouw 2023): if impossibility were basic, negation should outperform modal degree modifiers. Instead, *kaum* significantly outperforms pure negation ($\beta = 0.50, z = 5.48, p < .001$), showing that *gehen* tracks scalar potential rather than absolute impossibility.

This scalar requirement also shapes the lexical-semantic restrictions. Controlling for adverbial licensing in a Linear Mixed-Effects Model revealed a Vendlerian hierarchy (Vendler 1957): lexical Accomplishments with an inherent result state are the most productive predicates. Since middles characteristically ascribe a disposition to the promoted Theme and prefer Accomplishments/Activities over Achievements (§2.3), [*gehen+zu*-INFINITIVE] shows the core profile of a medio-passive strategy.

The strict rejection of Achievements follows from the same requirement. Achievements denote punctual events and encode a non-gradable two-point scale (Vendler 1957, 1967, Beavers 2008, Rappaport Hovav 2008), which cannot be mapped onto the durative scalar potential required by dispositional *gehen*.

The same scalar requirement explains the marginality of Activity verbs and compositional Accomplishments. Manner roots are only marginally tolerated under suitable adverbial modification, because atelic predicates can sometimes be coerced into a bound, gradable reading. Yet *noch* was too weak to force this coercion with *essen*, even though a quantized object can make 'eat the/an apple' telic by measuring out the event (see Tenny 1994). The data therefore distinguish inherent Accomplishments from Activity predicates that become telic only compositionally.

The remaining question is how the grammar enforces these restrictions, since they cannot be reduced to simple C- or S-selection.

4.3 Argument structure: Deriving syntactic and semantic restrictions

We now derive the argument-structural restrictions in the complement domain.

Properties

As established above, the rejection of agentive *von*-phrases shows that *gehen* does not project a syntactic AGENT argument. This aligns it with canonical reflexive middles (Steinbach 2002, Lekakou 2005, Schäfer 2008) and distinguishes it from *werden*-passives and *lassen*-middles, which freely license explicit agents. The comparative acceptability of *für*-phrases in [*gehen*+*zu*-INFINITIVE] is best read as evidence for an entailed or modally accessible participant, not for a syntactically projected Agent (see Stroik 1992, Ackema & Schoorlemmer 1995, 2015, Lekakou 2005, Pitteroff 2014). This interpretation fits the facilitative adverbial readings and the instrumental adjunct, which provide the stronger evidence for semantic agentivity.

Post-hoc qualitative diagnostics corroborate the absence of a syntactically active Agent (Roberts 1987, Roeper 1987, Baker et al. 1989, Bhatt & Pancheva 2006, Alexiadou et al. 2015): *gehen* prohibits both agent-oriented adverbs and control into *um* ...*zu* purpose clauses (39).

(39) a. **Das Spiel geht sorgfältig zu installieren.*
The game goes carefully to install
Intended: 'The game can be carefully installed.'

b. **Das Spiel geht einfach zu installieren, um Zeit zu sparen.*
The game goes simply to install in order time to
Intended: 'The game can be installed simply, in order to save time.'

If an Agent were structurally present, the sentences in (39) should be grammatical. Their failure indicates that both the infinitival *v*P domain and the verbal complex housing *gehen* are voiceless, lacking a projection capable of hosting a silent syntactic Agent.

The data also show that *gehen* is more restrictive than modal [*sein*+*zu*-INFINITIVE] in the internal-argument configurations it permits. The monotransitive dative item with *helfen* ('to help') is not decisive on its own, since *helfen* is an Activity. Nor is dative preservation a categorical middle/passive diagnostic, since German middles can preserve datives in special configurations, including impersonal *es*-middles (Steinbach 2002). The relevant contrast is narrower: modal [*sein*+*zu*-INFINITIVE] permits the dative and ditransitive configurations tested here, while *gehen* rejects them. We therefore treat these data as evidence for an additional selectional restriction on the complement of *gehen*, not as evidence for a general anti-dative property of middle voice.

The ditransitive item is structurally more informative: since *schicken* 'send' is an Accomplishment, its rejection cannot be reduced to the Activity confound affecting *helfen*. The problem lies in the configuration introduced by the second internal argument, namely the incompatibility between applicative structure and the ResultP-compatible complement selected by *gehen*.

Restrictions on Selection

LRIs in long passives select an agentless verbal complex, i.e., a specifierless VoiceP or, in earlier work, bare VP (Wurmbrand 2001, 2015). But this cannot simply be extended to *gehen*: ditransitives can appear inside such stripped-down infinitival domains, as scrambling from extraposed infinitives shows (see Wurmbrand 1998: 116). In (40), the accusative Theme of ditransitive *übermitteln* 'to send/transmit' scrambles into the matrix clause, stranding the dative goal.

(40) *weil der Hans [einen Brief]$_{SCR}$ versucht hat [der Maria t_{SCR}*
since the John [a letter].ACC tried has [to Mary.DAT
zu übermitteln].
to send]
'since John tried to send a letter to Mary'
(ex. 18 Wurmbrand 1998: 116)

On Wurmbrand's (2015) updated theory, which uses specifierless expletive VoiceP rather than truncation (contra Wurmbrand 1998, 2001), both low and high applicatives under VoiceP (Pylkkänen 2008) are predicted to be possible. Consultants accept an equivalent LRI containing scrambling, extraposition, and a high applicative (41), although such structures are degraded by their complexity.

(41) *weil der Hans [ein Haus]$_{SCR}$ versucht hat [seinen Eltern*
since the John [a house].ACC tried has [his parents].DAT
t_{SCR} zu bauen].
tSCR to build
'since John tried to build a house for his parents'

Returning to [*gehen*+*zu*-INFINITIVE], the diagnostics show that *gehen* cannot select an agentive VoiceP. Yet the absence of Voice does not by itself exclude applicatives: if [*gehen*+*zu*-INFINITIVE] selects only a *v*P,[9] high applicatives are ruled out, but low applicatives remain available. Recall the rejected ditransitive example in (42).

(42) a. **Die Email geht ihm nicht zu schicken.*
the email goes him.DAT not to send
Intended 'The email cant be sent to him.'

The ditransitive *schicken* 'to send' is both a low applicative (Pylkkänen 2008) and an Accomplishment, so it cannot fail for lexical-semantic reasons alone. The relevant restriction is that low applicatives are incompatible with resultatives (Pylkkänen 2008: 40–1): low-applicative arguments cannot relate to the subject of a small clause/ResultP.

(43) a. He painted me this flower. Low Applicative
b. He painted this flower blue. Resultative
c. *He painted me this flower blue. *Combination
(ex.69 Pylkkänen 2008: 40-1)

Therefore, we can posit the following restriction on *gehen*:

(44) **A selectional restriction on *gehen***:
The functional raising head *gehen* selects only for a resultative verbal complex involving a voiceless *v*P complement and ResultP.

How can a functional raising head impose such aspectual requirements on its *v*P complement? A C-selectional rule requiring a resultative v_{CAUSE} plus ResultP (cf. von Stechow 1995, Folli & Harley 2005, 2007, Alexiadou, Anagnostopoulou & Schäfer 2006b, Alexiadou, Anagnostopoulou & Schäfer 2006a) would not distinguish Accomplishments from Achievements, and standard 'flavors of *v*' analyses often treat v_{CAUSE} as agentive (see e.g., Folli & Harley 2005), forcing a special defective unaccusative v_{CAUSE} (but see Pylkkänen 2008). We therefore adopt the configurational approach of Alexiadou et al. (2015), where causative semantics are not encoded as narrow-

9 We treat *v*P as simply the introducer of the event and the verbaliser (Kratzer 1996, Embick 2004a,b)

syntactic head features.[10] Causation is instead interpreted at C/I whenever the syntax forms a "telic pair" $<e_1, e_2>$ (see Higginbotham 2000).

Beyond Selection

In the Distributed Morphology architecture assumed by Alexiadou et al. (2015), Narrow Syntax is blind to encyclopedic root content (Halle & Marantz 1993). Roots therefore do not enter the syntax as Accomplishments or Activities; these classes arise configurationally and are interpreted at C/I. If the syntax fails to derive the required bi-eventive configuration (*v*P dominating ResultP), or if the inserted root is conceptually incompatible with *gehen*'s dispositional requirements, the derivation crashes at the interface.

We assume that *gehen* C-selects an eventive *v*P but forbids an agentive Voice layer (see Kratzer 1996, Alexiadou et al. 2015). To satisfy the Accomplishment requirement, this *v*P must contain an eventive *v* head merged with ResultP; other configurations are filtered out at the interfaces.

The restrictions are enforced at LF and C/I. LF requires the *v*-to-ResultP configuration to compute the transition associated with a gradable scale. C/I requires a conceptual external force, i.e., an implicit Agent, because a Theme can only facilitate or impede an event relative to a potential actor (cf. Condoravdi 1989, Tenny 1994, Beavers 2008, Ackema & Schoorlemmer 1994).

Activities and compositional Accomplishments fail because their eventive *v* does not project ResultP. In a compositional Accomplishment such as *einen Apfel essen* ('to eat an apple'), telicity is derived by an event-bounding InnerAspP below *v* and above the root (Vendler 1957, Tenny 1994, MacDonald 2008, Travis 2010) (see 45).[11] InnerAspP measures out the event but does not introduce a distinct target state (s). When modal *gehen* probes its complement, it finds an event boundary but no result-state variable, so the dispositional relation cannot be established.

(45) [*v*P [InnerAspP [VP ...]]]

However, we must comment on apparent exceptions to the rule involving the atelic root *spiel* ('play') in (46).

10 This move is not a notational variant of a Vendler-class diacritic. On the decompositional syntax of event structure assumed here (*inter alia* Ramchand 2008, Travis 2010, Borer 2005, Alexiadou et al. 2015), aktionsart classes are *derived* from structure rather than listed: the same root may behave differently depending on whether an InnerAspP is forced (the *spielen*/*essen* contrast below), and the incompatibility of low applicatives with a result state follows configurationally (Pylkkänen 2008: 40–1).

11 See also Borer (2005) for a similar but crucially exoskeletal approach.

(46) *Jimmys Gitarre geht leicht zu spielen.*
Jimmy's guitar goes easy to play.INF
Intended: 'Jimmy's guitar plays easily.'

The acceptability of apparently atelic √SPIEL- ('play')[12] in (46) instead reflects structural coercion. Verbs of consumption involve an incremental Theme, forcing InnerAspP and starving *gehen* of a target state. An instrument like a guitar is not incremental, so the syntax need not build an aspectual measuring node and can merge the root in a resultative configuration. At C/I, *gehen* then locates the ResultP target state: the guitar is physically manipulated from an inert and silent state into an active and loud one. Thus atelic roots remain compatible with *gehen* when no incremental Theme forces InnerAspP, permitting coercion into a resultative structure.

Roots licensed by *gehen* appear to be either externally caused (e.g., √REPAIR, √INSTALL) or cause-unspecified roots permitting a durative process (e.g., √OPEN) (see the framework by Alexiadou et al. 2015).[13] Both are compatible with an entailed implicit Agent at C/I (see Condoravdi 1989, Schäfer 2008). Internally caused or spontaneous roots (e.g., √ROT) create a conceptual mismatch with external control, while punctual Achievement roots (e.g., √FIND, √BREAK) lack the durative process needed for gradable scalar evaluation.

Finally, because *gehen* neither selects nor is dominated by VoiceP, there is no syntactically active implicit agent; hence the ungrammaticality of *von*-phrases, agent-oriented adverbs, and control into purpose clauses. Low applicatives likewise fail because they block the required ResultP (Pylkkänen 2008: 40-41). Together, these constraints show that *gehen* is a middle raising head that forces detransitivization via unaccusative syntax (see also Abraham 2020): it merges without VoiceP and selects a bare *v*P/ResultP configuration, lacking the Voice architecture of canonical passives and active causatives (Alexiadou et al. 2015).

Let us now turn to set out a syntactic analysis of the entire derivation.

5 Analysis: From CP to ResultP

We established in the previous section that [*gehen*+*zu*-INFINITIVE] is a highly restrictive infinitival middle. Likewise, in Section 4.3, we already proposed

12 We use √ for the acategorial root of Distributed Morphology (Halle & Marantz 1993), the lexical core prior to categorization by *v*.

13 For the conceptual ontology of roots and the distinction between externally caused, internally caused, and cause-unspecified eventualities, see Levin & Rappaport Hovav (1995) and Levin & Rappaport Hovav (2005).

a theoretical framework with which to analyse the lexical *v*P domain under *gehen* and derive argument-structural and lexical semantic restrictions. We now present a step-by-step derivation, using (47) as an example derivation.

(47) ...*dass das Fahrrad leicht zu reparieren geht.*
...that the bike.NOM easily to repair goes
'...that the bike is easy to repair.'

5.1 *The vP domain: a recap*

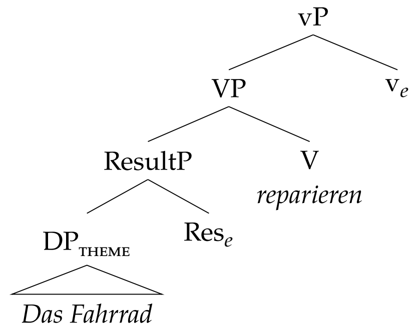
We have established that [*gehen*+*zu*-INFINITIVE] fails all tests of syntactic agentivity. We thus posit that *gehen* behaves like a functional raising verb which selects an eventive *v*P, forcing an unaccusative structure. The infinitival *v*P merges without a dominating agentive voice layer and thus represents a truncation of an otherwise canonically agentive causative predicate (*pace* Wurmbrand 2001).

We show the derivation of the infinitival complex in (48). We assume that the lexical predicate is built from an acategorial root introduced under V and verbalized by a categorizing *v*, which we label v_e to mark its eventive status (cf. Embick 2004a, Alexiadou et al. 2015). Crucially, German lacks V-to-*v* raising, so the lexical verb is spelled out in situ under V (Fuß 2008).[14] The reasons offered for the absence of V-to-*v* vary (a [−V-to-*v*] parameter (Broekhuis 2023), post-syntactic Morphological Merger under PF-adjacency in a head-final *v*P (Embick & Noyer 2001), the absence of the functional shells that would trigger raising (Haider 2005), and the lack of lower verb movement diagnosed by VP-topicalization (Hein 2021)), but they converge on a verb that does not raise to *v*, and we take no side among them. Both the lexical verb and, below, *gehen* itself are therefore introduced under V and categorized by an active *v*, with no head-movement between them. Moreover, recall the restriction for compatible Accomplishments to appear in inherently resultative configurations, e.g., change-of-state verbs such as *reparieren* 'to repair', as opposed to compositional Accomplishments, e.g., *essen* 'to eat' which comprise manner root with a quantised incremental Theme. We argue for a restriction for bi-eventive results that lexicalize a gradable scale. Descriptively, this aligns with a well-attested requirement for causatives to take a small-clause ResultP complement (cf. Hale & Keyser 1993, Von Stechow 1996, Hale & Keyser 2002, Folli & Harley 2005, Harley 2005, Alexiadou et al. 2006b). ResultP, we argue, is headed by Res^0 which introduces the secondary event variable

14 The derivations throughout assume head-final verbal projections; nothing in the account is incompatible with an antisymmetric approach that derives head-final order by leftward movement.

$<e_2>$, structurally deriving the telic pair $<e_1, e_2>$, cf. Higginbotham (2000). This is necessary because the DP is an entity and not an event.[15]

(48)



5.2 *Adding zu*

One question is how *zu* enters the derivation and what its role is. In our analysis *zu* does not perform or license any functional or semantic operation. Our principal claim is that the entirely unaccusative syntax of the truncated *v*P provides a passive interpretation for free (see Abraham 2020). This claim differentiates [*gehen*+*zu*-INFINITIVE] from [*sein*+*zu*-INFINITIVE], the latter of which has been argued to employ *zu* as the passivizer. For Holl (2010) *zu* is a passivizing head, while for Haiden (2005) and (Haider 2010) it is a passivizing affix which absorbs/saturates the external θ-role. We do not adopt these analyses, since i) they are beyond the scope of [*gehen*+*zu*-INFINITIVE], and ii) it is not clear how a passivising *zu* is compatible within the adopted Voice typology. Moreover, treating *zu* as a passivizing element, be it as an operator or argument-absorber, introduces unnecessary architecture into our derivation. [*gehen*+*zu*-INFINITIVE] lacks any syntactic Agent to be absorbed, bound, or saturated. Hence, even if *zu* encodes this function, it is redundant. Our view thus aligns with Wurmbrand (2001) who considers *zu* to lack semantic content but raises the problem of how it enters the derivation, given that it should be ruled out in the numeration, on account of Chomsky's Principle of Full Interpretation, Chomsky (1995: 24ff), and prohibited from entering it during the derivation (i.e., the Inclusiveness Condition, Chomsky 1995). We thus illustrate an assumed structure for the *zu* + V construct in (49).

15 Following standard event-structural analyses of Germanic separable particle verbs (den Dikken 2007, McIntyre 2004, Kratzer 2005), we take Res^0 to be the locus for the realization of verbal particles (cf. Ramchand 2008).

(49)

V^0

Inf V^0

zu *reparieren*

If correct, we must then account for the insertion of a formally inactive *zu*. We propose a PF-repair mechanism, assuming that vacuous items do not enter the derivation in the narrow syntax. Following Adger (2003) and Bjorkman (2011), the v head enters the derivation with an unvalued inflectional feature, [uInfl]. When the matrix T head values [uInfl] on a raising verb in the matrix clause, e.g., *gehen*, the embedded v head is isolated (see Wurmbrand 2001) and its [uInfl] remains unvalued. In order to avoid a crash at PF, the morphology must deploy a repair strategy. Following assumptions from Distributed Morphology (see Embick & Noyer 2001, Embick 2010, 2015), the system satisfies this requirement under the Elsewhere Condition (cf. Kiparsky 1973): *zu* is the default (elsewhere) infinitival exponent, inserted as a dissociated morpheme prefixed to v precisely when no more specific, valued [Infl] realization is available. Thus, the Inf node in (49) is the result of this structural repair at PF, i.e., it is "sprouted" post-syntactically (Halle & Marantz 1993, Halle 1997).[16] An anonymous reviewer observes that German verb clusters containing more than one non-finite verb appear to challenge this repair account: if an unvalued [uInfl] is realized as *zu*, the lower verb(s) of such a cluster ought spuriously to surface with *zu* as well. They do not, as the well-formed cluster in (50) shows.[17]

(50) ...*dass* *er* *das* *Buch* *hat* *lesen* *müssen*
that he.NOM the book has.3SG read.INF must.INF
'...that he has had to read the book'

We suggest, tentatively, that the resolution lies in how inflection is licensed across a coherent verbal complex. One avenue, inspired by the architecture Bjorkman (2011) develops for auxiliaries, treats inflectional licensing within such a complex as a chain of (reverse-)*Agree* relations: each verb carries an un-

16 Two clarifications. First, *zu* carries no independent value for [Infl]: it is the exponent inserted precisely *because* the isolated v's [uInfl] cannot be valued, not a goal that values it. Second, its linear position follows from its status as a proclitic dissociated morpheme left-adjoined to v, so it is realized immediately before the infinitive.

17 The converse pattern, where infinitival morphology surfaces *higher* than its canonical right-edge host, is found in the so-called *Skandalkonstruktion* (Reis 1979, Vogel 2009): in three-verb clusters with the *infinitivus pro participio*, the marker *zu* (and participial morphology) may be realized on a higher verb than expected. We take such non-canonical placement to be consistent with treating *zu* as a post-syntactically inserted dissociated morpheme whose linearization is fixed at PF rather than by a fixed syntactic head.

valued [uInfl] valued by the closest c-commanding inflectional source, and a verb left without any source spells out a default exponent: for Bjorkman a stranded feature surfaces as a default *auxiliary*, here as the affixal default *zu*.[18] On this view the members of a modal or IPP cluster are each licensed by the head immediately above them, the highest realizing finite morphology and the rest surfacing as bare infinitives, so that *zu* appears only on a *v* whose [uInfl] finds no source at all, namely the isolated complement of *gehen*. As a prevalued [iT] head, *gehen* is a closed goal for matrix T rather than a link in such a chain, and it selects a truncated, voiceless *v*P with no inflectional source of its own; the embedded *v* is the one position licensing never reaches, transferring with its [uInfl] unvalued and spelled out, as before, by *zu*. This isolates [*gehen*+*zu*-INFINITIVE] (with [*sein*+*zu*-INFINITIVE] and *scheinen*+*zu*) from the inflectionally integrated, bare-infinitive clusters of modals, perception verbs, and future *werden*.

We present this as a sketch rather than a settled analysis. Bjorkman (2011: 57) restricts her account to chains of two inflectional features, so extending it to longer German clusters (and to an affixal rather than a verbal default) raises questions we cannot resolve here. An alternative we leave open treats the coherent cluster as a single complex head in the spirit of Haider (2010), on which the lower verbs are not independent targets for inflectional realization, so that the prospect of a spurious *zu* does not arise in the first place. What we take to be secure is the empirical contrast between the integrated clusters above and the [*gehen*+*zu*-INFINITIVE] /*scheinen*/[*sein*+*zu*-INFINITIVE] type; the precise mechanism by which *zu* is realized in multi-verb clusters we leave to future work. We note a conceptual similarity to the PF-repair approach of Wurmbrand (2015), on which *zu* is the default reflex of a dedicated but deficient restructuring *v*; our account avoids that added architecture. In treating *zu* as a prefix on the infinitive we join Demske-Neumann (1994), Haider (2010: 349–350), Abraham (2020), and Haiden (2005).

5.3 *Adding gehen*

Now *gehen* must combine with the lexical predicate. As established *gehen* behaves like a functional restructuring verb, lacking theta-assigning capabilities

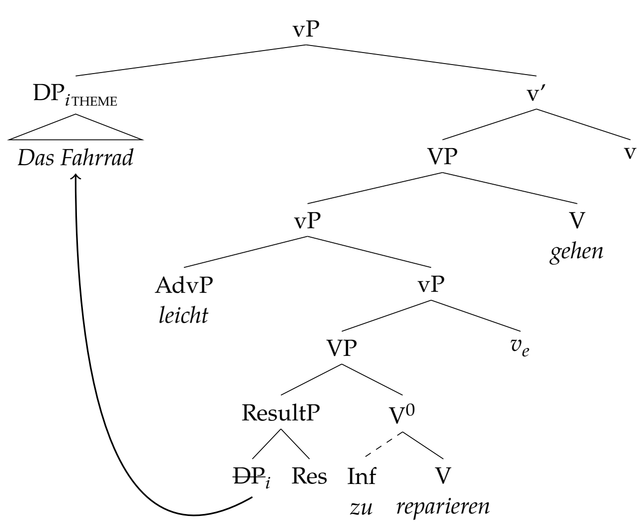
18 Cluster-internal licensing under a closest-c-command minimality condition has independent precedents in feature-driven cluster formation. Sabel (2001) derives multiple head movement, clitic climbing, and multiple wh- and focus fronting through a single [+affix]-driven attraction mechanism constrained by c-command closeness; Grewendorf (2001) derives multiple wh-fronting as wh-cluster formation under the same minimality; and Grewendorf & Sabel (1999) treat German and Japanese scrambling as feature-driven movement to multiple specifiers (versus adjunction). These accounts derive the formation and ordering of moved elements but do not invoke the default-exponent mechanism we adopt here for stranded [uInfl].

and inducing Raising. However, it is more restrictive than *scheinen* 'seem' regarding the nature of the selected complement. This property means that *gehen* appears to retain some lexical content like a semi-functional head, yet in fact its only selected restriction is for an eventive *v*P. We argued that the lexical-semantic restrictions are derived by the configurational properties of the predicate and its compatibility with *gehen* in deriving a gradable dispositional property assigned to the Theme. In contrast, *scheinen* does not place argument-structural or lexical-semantic constraints on its predicate, even selecting TPs.

Moreover, we treat functional raising verbs as unaccusative *v* heads (see Section 2.4) (*inter alios* Grewendorf 1989, Chomsky 2001, Davies & Dubinsky 2004, Haider 2010) and not a higher functional projection (*contra* Cinque 2006a, Wurmbrand 2001). This position is supported by our repair account of *zu*; if a raising verb occupied a higher projection, nothing prevents it from valuing [uInfl] on the lexical *v*, thus bleeding insertion of *zu* at PF. The puzzle is then why *gehen* and *scheinen* resist compound tense, and not why they force restructuring. Concretely, *gehen* too is introduced under V and categorized by this raising *v*: consistent with the lack of V-to-*v* in German, *gehen* is spelled out in V, while it is the categorizing *v* (not V) that carries the features driving the derivation (the valued [iT] and the structure-building edge feature introduced below). Calling *gehen* a raising *v* is thus shorthand for this V+*v* complex, exactly as the trees in (51) and (54) depict it: *gehen* in V, with [iT] on *v*.

The derivation for the entire raising verbal complex is given below (51).

(51)



We argue that the Theme moves to a Spec,*v*P position, since the standard analysis (based on English) dictates that unaccusative raising verbs are defective, non-phasal, and lack a specifier position (Chomsky 2001). In English, which has a subject-related EPP on T, T can then probe the embedded DP directly as probing is not blocked by the phase edge. This triggers subject raising to Spec,TP. However, German is not standardly assumed to have a subject EPP requirement on TP (Abraham 1993, Haider 1993, 1997, 2010, Sternefeld 2006, Biberauer & Roberts 2010).[19] Indeed for *scheinen*, there does not appear to be any issue as in-situ subjects are known to be unproblematic.

However, for [*gehen*+*zu*-INFINITIVE] there is good evidence that the DP Theme must raise to a position in the middle field; this is evident in the contrast in (52a,b) which shows [*zu*-INFINITIVE] can only be fronted without the DP theme, which must evacuate to middle field. Moreover, (53) strengthens this conclusion, demonstrating that the Theme must evacuate to a position higher than the evaluative adverb, which we consider a *v*P level modifier, but lower than CP. We assume this position to be Spec,*v*P, but admit that these data are also consistent with a Spec,TP analysis which we reject on wider theoretical grounds for German.

(52) a. *Zu öffnen gehen die Fenster leicht.*
to open goes the window easily
'The windows are easy to open.'

b. **Die Fenster zu öffnen gehen leicht.*
the windows to open goes easily

(53) *Damals ging (??/*schwer) ein Auto schwer zu reparieren.*
back_then went (difficult) a car difficult to repair
'Back then, a car was difficult to repair.'

We propose that the necessity for displacement is tied to the head as a *gehen*-middle is not a canonical raising construction; it is a dispositional middle.[20] Following proposals for category-flexible structure-building EPP features, e.g., [•X•] by Müller (2014), we argue that this *v* specifically bears a structure-building [•D•] feature (see also Sluckin 2021). The mandatory [•D•] feature on the raising *v* forces the Theme into the local predicational configuration with *gehen*, which is necessary to derive the dynamic dispositional property of the Theme. This analysis explicitly requires us to treat v_{gehen} as a phase head (Chomsky 2001, 2008), adopting *v* to always be a phase

19 This does not mean that Spec,TP never projects, but that probing from T cannot be relied on as the mechanism for subject raising; it is undesirable to reduce Raising to a scrambling-only strategy.

20 We refer to the discussion of micro and nanoparameters by Biberauer & Roberts (2017).

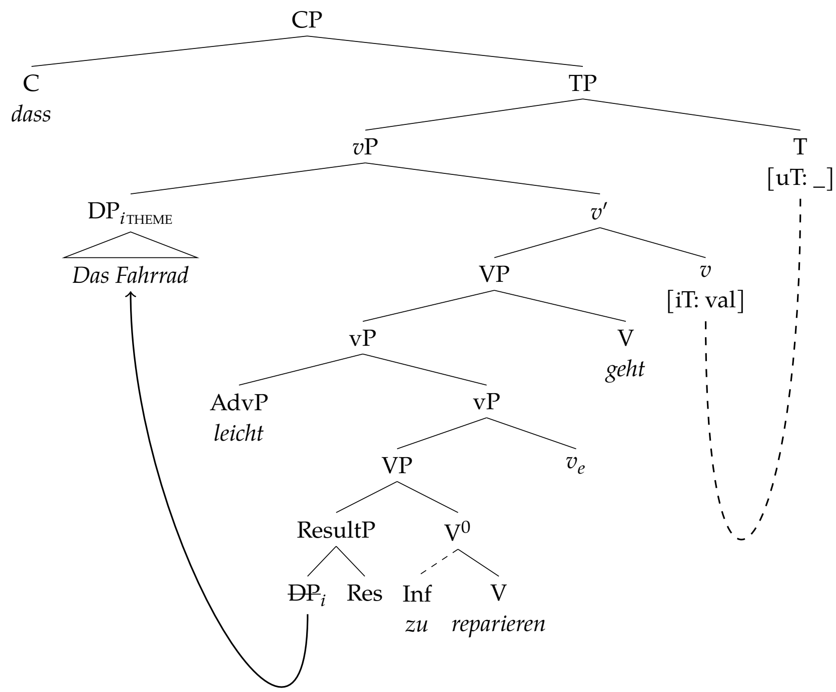
(Arad 2003, Marantz 2001, 2007, den Dikken 2007, Legate 2003, 2014). Treating *gehen* as a phase head also supports the account of *zu* insertion. Namely, once the Theme lands in Spec,*v*P, the *v*P phase is complete, and v_{gehen} and its complement, the embedded *v*P, will undergo transfer to the interfaces (Chomsky 2001, 2008). Subsequently, when the embedded *v* head with unvalued [uInfl] is sent to PF, the repair strategy is triggered that inserts the dissociated morpheme *zu*.

5.4 *The functional domain: bleeding auxiliaries*

The functional domain, specifically the relationship between matrix T and the raising *v*, requires separate treatment. Recall that both *gehen* and *scheinen* totally resist compound tenses such as the perfect. Therefore, we propose a system by which combination of *gehen* with compound past tense morphology is ruled out at the syntax-morphology interface.

To account for the strict incompatibility of [*gehen*+*zu*-INFINITIVE] with compound tenses, we adopt a feature-valuation based approach (cf. Chomsky 2001), assuming that functional raising verbs of this type enter the derivation fully specified for tense (cf. Wurmbrand 2001). Specifically, the functional head *gehen* is merged bearing the valued interpretable feature [iT]. The higher T head enters the derivation bearing unvalued, uninterpretable features [uT:_]. In a standard probe-goal configuration, T probes its c-command domain and locates the matrix v_{RAISING} head hosting *gehen* (Chomsky 2000, 2001). The unvalued features on T automatically enter into an *Agree* relation with this locally c-commanded goal, receiving their values directly from the [iT] features on *gehen*. Crucially, because this long-distance *Agree* relation successfully values T's features in-situ, it bleeds the generation of a higher temporal auxiliary. Following Bjorkman (2011), temporal auxiliaries are semantically vacuous default verbs, inserted as a last resort to realize *stranded* inflectional features (those an intervening head keeps from combining with the main verb under *Agree*); in her terms, auxiliaries have an 'overflow' distribution, not being tied to a single inflectional category. As a result of the raising head valuing T, no features remain that could license or trigger Merge of an auxiliary. The derivation of a full clause is given below (54).

(54)



This bleeding effect is specific to *temporal* auxiliaries. For Bjorkman (2011), perfect and tense auxiliaries are exactly this kind of last-resort default: semantically vacuous verbs merged only to realize inflection that would otherwise be stranded. Once *gehen* values T in situ, no stranded feature remains and the perfect auxiliary is correctly barred. A finite *modal*, however, is not such a last-resort default: it is a contentful head, merged for its own modal semantics, and is independently the finite, tense-bearing verb of its clause. Accordingly (and unlike the perfect), *gehen* is compatible with a superordinate finite modal, where it surfaces as a bare infinitive rather than a finite form (55).

(55) a. *Jetzt, wo das Gebüsch kahl ist, müsste die Sache doch*
now where the bush bare is must.SBJV.3SG the thing PRT
gut zu reparieren gehen.
well to repair go.INF
'Now that the bushes are bare, the thing really ought to be easy to repair.'[21]

b. *Sollte zu beheben gehen!*
should.3SG to fix go.INF
'[It] should be fixable!'[22]

c. *muß* *doch* *zu* *schaffen* *gehen.*
must.3SG PRT to accomplish go.INF
'[It] really must be doable.'[23]

We take the modal to merge above the *gehen*-projection as a contentful (semi-)functional head that itself carries valued tense. As the structurally closest goal to matrix T, the modal values T's [uT] under *Agree* by minimality; *gehen* therefore neither needs to nor does value T in this configuration, remaining in situ and contributing only its dispositional semantics. Because the modal is licensed independently of inflectional repair, the last-resort logic that excludes the perfect auxiliary does not exclude it.[24] The relevant contrast is thus not between *gehen* and higher material *per se*, but between semantically vacuous tense auxiliaries, which *gehen* bleeds, and contentful modal heads, which it tolerates.[25]

5.5 *A final word on modality*

Finally, for *sein+zu* or *haben+zu* modal infinitives, Holl (2010) and Abraham (2020) propose that the modal content is derived from *zu* itself; Abraham (2020) derives modality configurationally from the directional preposition *zu*, while Holl (2010) proposes a homophonous modal-passive *zu*. In contrast, we have already proposed that *zu*, at least for [*gehen+zu*-INFINITIVE], is in fact the result of a repair strategy at PF. We have followed the position that *gehen* is a *v* head encoding dynamic possibility (see also Holl 2010). Holl (2010: 10) argues this based on the availability of a possibility reading in simple matrix clauses without a *zu*-infinitive (56).

(56) *Eine solche Übersetzung geht in zwanzig Minuten.*
a such translation goes in twenty minutes
'Such a translation is possible in twenty minutes.'
(ex.5d Holl 2010: 10)

However, our data also clearly show that modification via an evaluative adverb, a gradable degree adverb, or negation is required for grammaticality.

24 A reviewer notes that a finite modal embedding *gehen* can itself appear under a perfect auxiliary, e.g. *Eigentlich hätte das Fahrrad doch gut zu reparieren gehen müssen* 'Actually, the bike really should have been easy to repair'. This is not a counterexample. The perfect auxiliary here realizes the perfect of the *modal* (*müssen*, in its IPP form), not of *gehen*: the finite tense locus is *hätte*, and *gehen* still values no tense of its own. What the last-resort logic excludes is a perfect auxiliary taking *gehen* directly; it does not exclude a perfect auxiliary stacked over a higher contentful modal that embeds *gehen*. The prediction is thus a contrast between an ill-formed direct perfect of *gehen* and the well-formed *hätte* … *gehen müssen*.

25 These are naturalistic web attestations; we leave a systematic study of *gehen* under modals to future work.

These two conclusions contradict each other; if *gehen* were the pure source of dynamic possibility, a dispositional reading should be available without further modification.

Let us then consider the feature specification of *gehen*. We assume that *gehen* enters the syntactic derivation bearing a valued, interpretable modal feature [iMod]. At C/I, this feature is interpreted as conveying existential modal force (◊) evaluated against a circumstantial modal base which is anchored to the physical properties of the internal argument (see Kratzer 1991, Lekakou 2005, Holl 2010, Pitteroff 2014, Abraham 2020). Since *gehen* selects a bare *v*P complement lacking a syntactically active agent, an interpretation relying on a deontic ordering source, which requires an entity subject to laws or obligations, is excluded *a priori* (cf. Bhatt 2006). Instead, the pragmatic context provides a stereotypical ordering source that represents the normal progression of events based on the physical properties of the Theme. However, although *gehen* comes preloaded with this modality, it is too weak to yield the necessary dynamic dispositional reading on its own. To elicit the dispositional reading, the modality must evaluate the promoted Theme's potential to facilitate the event. This requires a gradable scale to measure exactly how easily that change of state can be achieved. Consequently, the addition of an appropriate adverbial modifier provides the scalar infrastructure needed for the dispositional relation present in dynamic modality, explicitly anchoring this evaluation to the properties of the Theme.

Crucially, our proposals correctly predict the non-dispositional behaviour of *gehen* in control structures. As shown in (57), the dynamic dispositional reading vanishes when *gehen* takes a clausal subject, whether extraposed as in (57a) or fronted as in (57b).

(57) a. *Es geht nicht,* [PRO$_{arb}$ *den Wagen zu reparieren*].
it goes not the.ACC car to repair
'It is not possible/not permitted to repair the car.'

b. [PRO$_{arb}$ *einen Wagen kaputt zu machen*] *geht gar nicht.*
a car broken to make goes at-all not
'It is absolutely unacceptable to break a car.'

In both configurations, the internal argument remains trapped within the embedded Case domain. Because the Theme does not undergo A-movement into the matrix clause, it cannot establish a dispositional relation with *gehen*. Instead, these clausal subjects project a PRO$_{\text{arb}}$ in the active thematic VoiceP dominating the infinitival *v*P. This structurally provides a higher entity capable of bearing obligation or permission. The existential operator is evaluated against a circumstantial modal base, while the pragmatic context determines the flavour of its ordering source. For example, when evaluated against a deontic (normative) ordering source, it yields a strong social prohibition reading in (57b). Although fundamentally different at the structural level, this proposal resembles then the logic of Holl's (2010) Kratzerian argumentation for variation in the modal interpretation of *zu* in [*sein*+*zu*-INFINITIVE], yet we derive the variation from a combination of the preloaded modal content of *gehen*, the relevant pragmatic context, and the nature of the operation, e.g., Raising vs Control, which is ultimately dictated by the argument structure of the embedded predicate and its thematic properties.

Finally, the arbitrary/generic flavour of the *gehen*-middle must be accounted for, especially given the absence of the silent generic operator (Gen) typically posited for canonical middles (Lekakou 2005, Schäfer 2008). We propose that a generic reading emerges as a byproduct of the lack of Voice and its modal semantics. A thematic Voice head links the Agent to the specific episodic event via Event Identification (Kratzer 1996). However, this is not possible as *gehen* selects a *v*P. Consequently, the agentive semantics entailed by the encyclopedic knowledge of the root cannot be mapped to a specific event. This unmapped Agent interacts directly with the existential modal operator introduced by *gehen*. The modal operator can only evaluate the dispositional feasibility of the target state, rather than anchoring an event to a syntactically encoded Agent. Therefore, the unmapped implicit Agent receives an arbitrary default interpretation. Thus, the generic flavour is an interpretive consequence of scoping a modal operator over an unaccusative configuration built from an encyclopedically agentive root.

The *gehen*-middle is a tightly restricted configuration, defined by a constellation of interacting formal properties. Given that German already possesses a productive canonical middle, the emergence of this idiosyncratic periphrasis presents a historical puzzle. The construction bears a notable similarity to modal control structures under *gehen* (cf. Holl 2010), which we assume to be its source. If correct, we must ask how and why a biclausal control structure was reanalysed as monoclausal raising structure.

6 The historical development of the *gehen*-middle

Previous work on the historical development of non-canonical passives in German has mostly focused on modal infinitives, i.e., [*sein*+*zu*-INFINITIVE] (Behaghel 1923, Ebert 1976, Boon 1981, Demske-Neumann 1994). Eroms (1990) offers brief remarks on other construction types that somewhat resemble the regular passive, but to date, no diachronic studies have dealt explicitly with the *gehen*-middle. We attempt to narrow this gap, focusing on the following research questions:

i. What is the historical origin of the *gehen*-middle (i.e., its earliest attestations and potential sources)?

ii. How has the construction developed subsequently with regard to its distribution across verb classes, text types, and syntactic contexts (root vs. embedded clauses)?

iii. Which factors governed its emergence and further development?

In connection with the first research question, we would like to pursue the hypothesis that the *gehen*-middle evolved from a reanalysis of (arbitrary) control structures linked to the expression of possibility (after *gehen* had developed a relevant meaning):

(58) possibility reading of 'to go' → control structures → raising structures (structural simplification)

Before we flesh out this proposal in more detail, we take a closer look at the relevant diachronic facts, presenting the results of a corpus study carried out in the historical corpora of the *Digitales Wörterbuch der Deutschen Sprache* (DWDS).

6.1 *The rise and loss of the gehen-middle – a corpus study*

To trace the historical development of the *gehen*-middle, we conducted a corpus study in the historical subcorpora of the DWDS, which cover a time span from 1465 to 1987 and contain approximately 294 million tokens.[26] We extracted all cases (i) where an infinitive (VVINF/VVIZU) precedes a form of 'to go' and (ii) where a form of 'to go' precedes an infinitive (max. distance 5 words, with the infinitive at the end of the clause).[27]

The search produced 3900 hits, which were narrowed down manually to 95 relevant examples, spanning the years 1715 to 1931. Interestingly, the num-

26 We used the DWDS corpora, which cover the early and late Early New High German/early New High German period when the *gehen*-middle likely emerged; searches in the reference corpora of Middle and Early New High German and the GermanC corpus yielded no relevant results.

27 The two search patterns are deliberately asymmetric. Pattern (i), with the infinitive preceding finite 'go', targets verb-final (hence embedded) clauses, where the two are string-adjacent by independent rule, so no distance limit is imposed. Pattern (ii), with 'go' preceding the infinitive, targets V2/root clauses; here a wide window is counterproductive, because *gehen* is highly polysemous, and an unrestricted gap fills the results with motion-'go' followed by an unrelated purpose infinitive (*geht* ..., *um zu* ...). Limiting pattern (ii) to a clause-final infinitive within five words reproduces the adjacency profile of the genuine construction (in which at most an adverb or negation intervenes, as in *Das geht nicht zu ändern*), while screening out this purpose-clause noise.

ber of hits far exceeded those in present-day corpora such as DeReKo, despite the smaller size of the historical corpus. The earliest examples found date from the early eighteenth century, given in (59).

(59) a.

Heisset	*ein*	[...]	*Tüchlein,*	*brennend*	*in*	*das*	*Wein-Faß*	
bid	a	[...]	handkerchief	burning	into	the	wine-barrel	
hängen,	*welches*	*zu*	*zapffen*	*gehet,*	*oder*	*nicht*	*voll*	*gefüllet*
hang	which.NOM	to	tap.INF	goes,	or	not	fully	filled
ist,	[...]							
is,	[...]							

'Put a burning handkerchief in the wine barrel, which can be tapped or is not entirely full'
(Corvinus, Gottlieb Siegmund: Nutzbares, galantes und curiöses Frauenzimmer-Lexicon. Leipzig, 1715)

b.

so	*bald*	*nun*	*Adversarius*	*pariret*	*haben*	*wird*	*und*	*von*	*der*
as	soon	now	adversary	parried	have	will	and	from	the
Kling	*die*	*Quart*	*zu*	*stossen*	*gehe*	*drehet*	*man*	*den*	
blade	the	quarte.NOM	to	thrust.INF	goes	turn	one	the	
Leib	*in*	*Geschwindigkeit*	*herum*	[...]					
body	in	pace	around	[...]					

'As soon as the opponent has parried and the quarte can be thrusted with the blade, the body should be turned with pace...'
(Doyle, Alexander: Neu Alamodische Ritterliche Fecht- und Schirm-Kunst. Nürnberg u. a., 1715)

As can be seen from Figure 8, there is a gap of around 100 years between the above examples and later cases. Throughout the nineteenth century, the construction remains a low frequency phenomenon (< 0.5 occurrences per million tokens).[28] The bulk of all cases comes from the (late) nineteenth century; thereafter, the construction seems to be on its way out.

Taking a closer look at the syntactic contexts in which the *gehen*-middle occurs, it turns out that the construction is not evenly distributed over clause types and verb classes, revealing some similarities but also differences between the historical cases and present-day instantiations of the *gehen*-middle. First, and similar to present-day German (PDG), we can observe an effect of verb class. The majority of all cases involve Accomplishments (69/95 cases, 72.6%). A typical example involving the verb 'to change' is shown in (60).

28 Note that the normalized frequencies for the eighteenth century are misleading – there are only two examples, but the amount of text is much smaller than in the (late) nineteenth century; basically the same goes for the years after 1918.

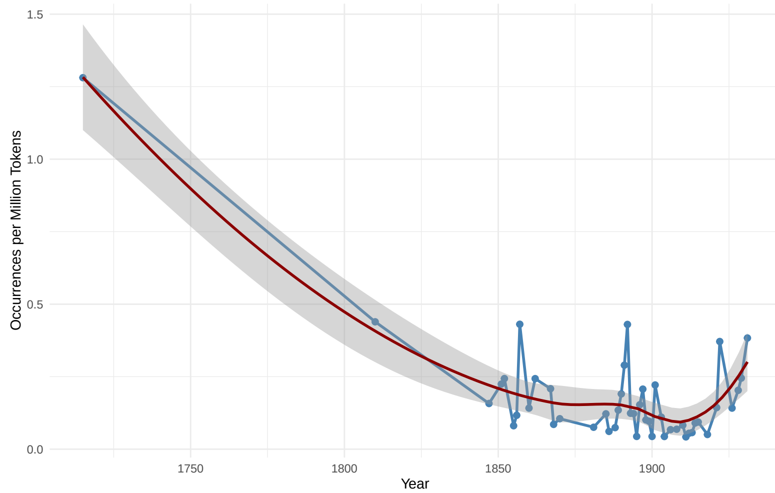


Figure 8 Historical Development of the *gehen*-middle (normalized frequencies per year and Loess-smoothed curve)

(60) ***gehen* + Accomplishment**

Was	*einmal*	*geschehen,*	*das*	*geht*	*nicht*	*zu*	*ändern.*
what	once	happened	that.NOM	goes	not	to	change.INF

'What happened once cannot be changed.'
(Deutsches Sprichwörter-Lexikon. vol. 1, 1867, p. 1585)

However, there is also more variation than in PDG, including Achievements as in (61) (20/95 cases, 21.1%) and Activities as in (59a) above (6/95 cases, 6.3%), see Table 3 and Figure 9.

(61) ***gehen* + Achievement**

der	*[...]*	*Marketender*	*versorgte*	*uns*	*mit*	*Allem,*	*was*	*nur*
the	[...]	sutler	provided	us	with	everything	what.NOM	only

zu	*kaufen*	*geht*
to	buy.INF	goes

'The sutler provided us with everything that could be bought.'
(Stolle, Ferdinand (Hrsg.); Diezmann, August (Hrsg.): Die Gartenlaube. Jg. 4, 1856)

Clause type	Accomplishment	Achievement	Activity	State	Total
Embedded	61 (74.4%)	16 (19.5%)	5 (6.1%)	0	82
Root	8 (61.5%)	4 (30.8%)	1 (7.7%)	0	13
Total	69 (72.6%)	20 (21.1%)	6 (6.3%)	0	95

Table 3 Historical instances of the *gehen*-middle – distribution over clause types and verb classes

Throughout the embedded cases, finite *gehen* sits immediately to the right of the *zu*-infinitive, and in no embedded instance is the infinitive stranded at a distance from 'go'. This adjacency, rather than the raw clause-type counts, is what we read as consistent with verb clustering as a precondition for the construction's licensing and grammaticalization, matching the coherence requirement of the synchronic analysis (Section 5). The skew toward embedded clauses (82/95 cases, 86%) corroborates this, but only that. Recall is not balanced across clause types: pattern (ii) imposes a clause-final, short-distance window that pattern (i) does not (see the query above), so genuinely root cases with material between 'go' and the infinitive (including the wider range of clause-final extraposition and longer embeddings that older German still permitted) fall outside the search. The embedded share is therefore partly a search artifact, and the root-clause adjacency is itself partly forced by the query; we accordingly attach no theoretical weight to the root/embedded ratio, resting the clustering claim on the absence of stranding instead. The verb-class generalization is untouched by this. The dominance of Accomplishments holds *within* both bands (Figure 9, which also shows that the earliest examples such as (59) are Activities rather than Accomplishments), so the aspectual conclusion does not depend on the clause-type distribution. Both halves of the study, the synchronic experiment and this first corpus survey of the construction's history, are exploratory; the diachronic proportions are offered as first-pass description rather than as load-bearing statistics.

Additional grammatical and extra-grammatical factors that influence the distribution of the construction in the historical data include tense, the presence of adverbs as well as negation, and genre/text type.

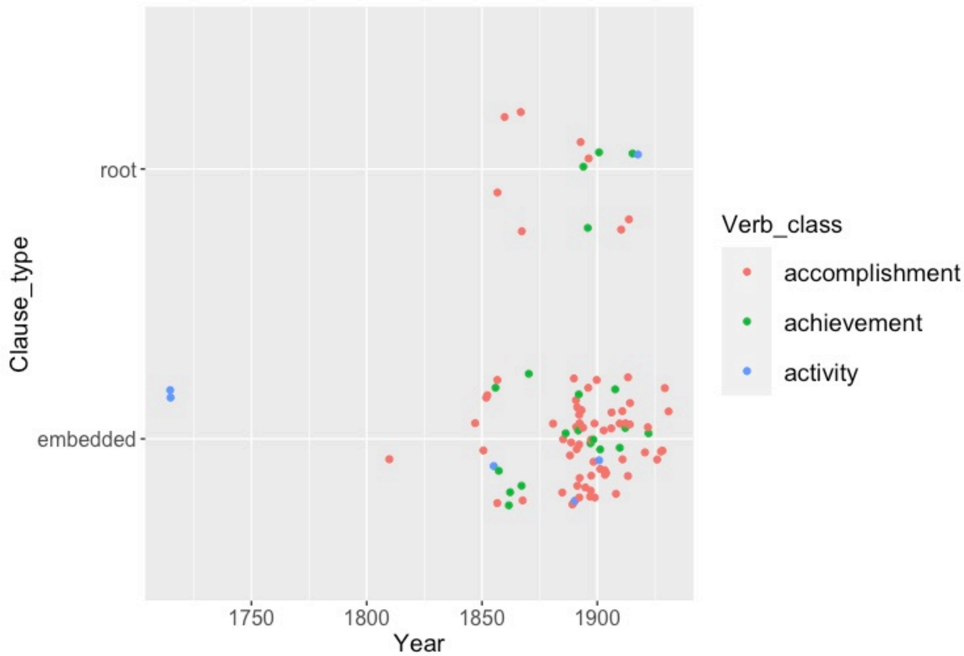


Figure 9 The 'go to' passive in the historical DWDS corpora: clause type and verb class. The vertical axis is categorical (root vs. embedded); points are jittered vertically for legibility, so vertical spread within a band is not meaningful.

Tense The vast majority of examples (73/95, 77%) are in the present tense, which is probably a reflex of the generic flavour of the construction. But again, there is more variation in the historical data. In contrast to PDG, the *gehen*-middle is not confined to simple tenses but also shows up with periphrastic verb forms as in (62).

(62) **periphrastic verb form**

er	*sei*	*aber*	*nicht*	*auszuheben*	*gegangen*
he.NOM	be.BE.PRES.SBJ	however	not	out-to-lift.INF	gone

'However, it [the shutter] couldn't be taken off its hinges'
(Die Ermordung der Witwe Dellbrück, geb. Hahnemann. In: Der neue Pitaval, Bd. 28. Leipzig, 1860)

Presence of adverbials/negation Modification by some kind of adverbial expression is the most common pattern (49/95 cases, 51.6%). Some relevant cases are shown in (63). The negation *nicht* 'not' is present in 19/95 cases

(20%), see e.g. (60) above for illustration. In all attested cases, negation scopes over *gehen*, yielding an impossibility reading ($\neg\Diamond$: 'cannot be V-ed'); this is consistent with the analysis in Section 5, where *gehen* occupies the matrix modal head position. There is no evidence in the corpus for narrow-scope negation ($\Diamond\neg$: 'it is possible to not V').

(63) a. *Da diese letzteren am leichtesten abzuändern gehen* [...]
since the latter.PL on-the easiest modify.INF go.PL
'Since the latter are the easiest to modify...'
(Theodor Pregél, Die Ermittelung der Stufenscheibenverhältnisse für Werkzeugmaschinen. In: Dinglers Polytechnisches Journal, 1893/287, pp. 248-252. Stuttgart, 1893)

b. *Daß es auch billiger zu machen geht, als die*
that it.NOM also cheaper to make.INF goes than the
Regierung vorschlägt, das beweist uns die Schweiz
government proposes that proves us the Switzerland
[...].

'That it can also be done more cheaply than the government proposes is demonstrated by Switzerland.'
(Verhandlungen des Reichstages. Berlin, 1899.)

However, a surprisingly large portion of the historical examples lack a modifying element as in (64) (27 cases, 28.4%), which is not expected given the PDG facts, where the presence of adverbials or negation seems to be compulsory.

(64) *die* [...] *einen Schwiegersohn wünschte, der zu*
who.FEM [...] a son-in-law wished who.NOM.MASC to
lenken und leiten [...] *ging*
steer.INF and lead.INF [...] went
'who wished for a son-in-law who could be steered and led'
(Keil, Ernst (Hrsg.): Die Gartenlaube. Jg. 3 (1855))

Impact of genre/text type I The vast majority of all occurrences appear in utility prose and science (subgroups according to the DWDS corpus; manually added: 'science' for papers in *Dinglers Polytechnisches Journal*, one of the foremost technical journals in nineteenth century Germany). The distribution over different genres is visualized in Figure 10 (absolute numbers of occurrences and normalized frequencies per million words).

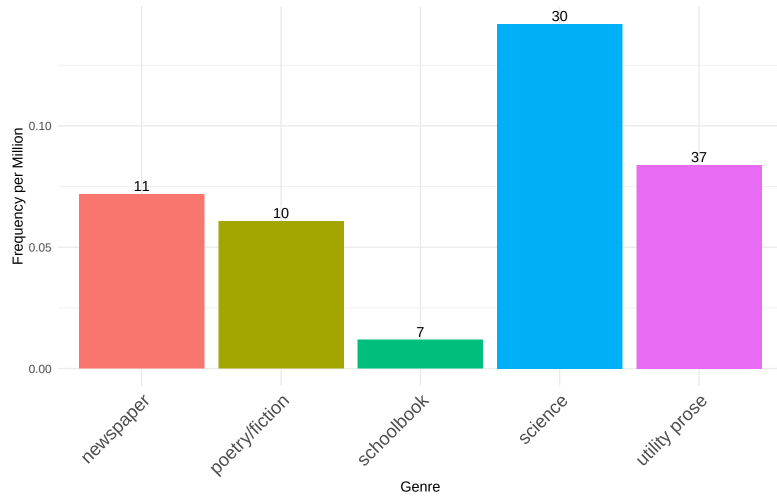


Figure 10 Distribution of the *gehen*-middle across text types (normalized frequencies and absolute numbers)

Impact of genre/text type II There is a striking spike in texts linked to the genre 'science' in the late nineteenth century, while examples in other types of utility prose are more evenly distributed: see Figure 11.

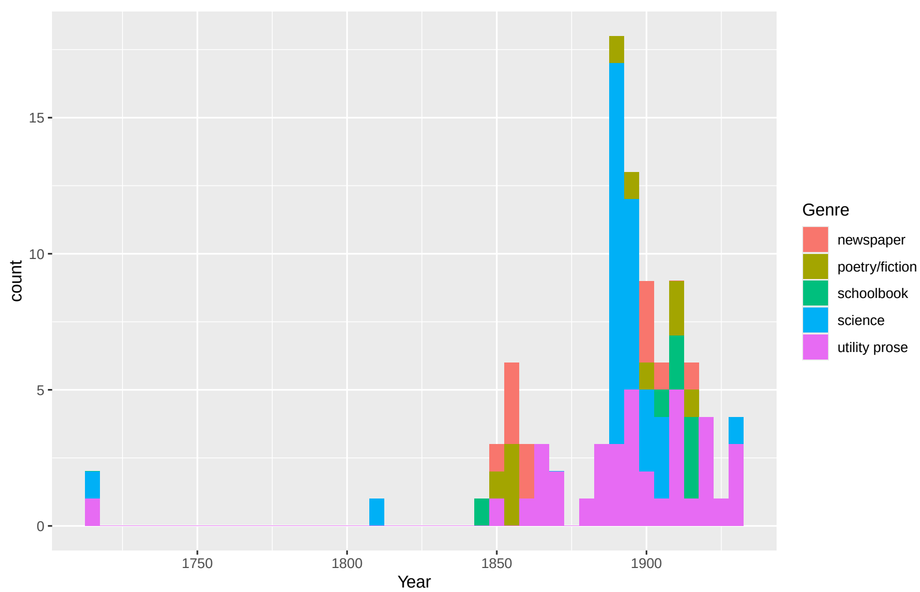


Figure 11 Proportion of text types per 10 years

All relevant examples in Figure 11 (27 cases) appeared in *Dinglers Polytechnisches Journal*. A typical example of this type of technical writing is given in (65).

(65) *so dass die mit äusserem Schraubengewinde versehene*
so that the with external screw-thread provided
Bohrspindel leicht einzubringen und herauszunehmen
drill-spindle.NOM easy to-insert.INF and to-remove.INF
geht
goes

'so that the drill spindle, which is provided with an external screw thread, can be easily inserted and removed'
(Neuerungen in der Tiefbohrtechnik von Eugen Gad in Darmstadt. In: Dinglers Polytechnisches Journal (Hg. A. Hollenberg), Jg. 1890/278, S. ad-156. Stuttgart, 1890)

However, upon closer inspection, the sudden increase could in fact reflect individual stylistic preferences, since at least 10 of all 27 cases appear in articles by Theodor Pregél, a professor of engineering at the University of Chemnitz, see (66) for another example reflecting this kind of style. Regardless, the use of the *gehen*-middle in text types such as science/technical writing is not expected, given its present-day link with colloquial speech.

(66) *An einer Ringleiste des Rundtisches ist der Schneckenkranz h*
at a ring-rail of-the rotary-table is the worm-gear h
angebracht, welcher durch die Schnecke i zu betreiben geht.
mounted which.NOM through the worm i to drive.INF goes

'At a ring rail of the rotary table, the worm gear h is mounted, which can be driven by the worm i.'
(Th. Pregél: J. E. Reinecker's Werkzeugmaschinen. In: Polytechnisches Journal (Hg. W. Pickersgill), Jg. 1901/316, S. 357-361. Stuttgart, 1901.)

The major empirical results of the corpus study are as follows: From the data we have gathered, the *gehen*-middle appears to have evolved rapidly between the eighteenth and nineteenth centuries (recall that the construction does not seem to exist in earlier historical stages of German). We have also noted a strong tendency for *gehen* to be right-adjacent to the infinitive, which suggests that verb cluster formation was a necessary precondition for the grammaticalization of the *gehen*-middle. The construction reached its prime in the late nineteenth century (in certain text types, technical writing, in particular); at that time, it had a wider distribution across verb classes, verb forms, and non-colloquial registers. Afterwards, the *gehen*-middle seems to be in decline, at least in the written language. Interestingly, the increasing restriction to colloquial registers appears to go hand in hand with growing grammatical constraints, a development that ultimately excluded periphrastic verb forms and Achievements from the modern construction. Both of these losses connect directly to the synchronic analysis. The exclusion of Achievements and Activities (still admitted in the historical data, Table 3) is the hardening of *gehen*'s requirement of a structurally distinct result state, modelled in Section 5 as the selection of a complement ResultP, from a statistical preference into a categorical selectional restriction. The loss of compound tense points the same way: the earlier compatibility of the *gehen*-middle with periphrastic forms (62) indicates that 'go' had not yet become the inflectionally self-sufficient raising head of Present-Day German, the head whose valued tense bleeds the last-resort perfect auxiliary (Section 5). On this view its loss is a late reflex of the same functionalization that completed the reanalysis into a raising structure, so the historical compound-tense cases do not undermine the synchronic claim but rather date the final step of the change.

6.2 *The origin of the gehen-middle*

In many languages, passive auxiliaries can be traced back to lexical verbs. Relevant processes of grammaticalization frequently affect verbs expressing inchoative or change-of-state meanings (Haspelmath 1990, Heine & Kuteva 2002, Wiemer 2011). Well-known examples include 'get/receive' (English, ancient Chinese), 'become' (German), and 'give' (Luxembourgish and Moselle Franconian dialects, Nübling 2006, Lenz 2007). Another likely source are motion verbs, in particular, 'come' (in the so-called 'Alpine passive', Wiesinger 1989, Ramat 1998, Gaeta 2018) and 'go' (Urdu, Quechua, Scottish Gaelic, Alpine and colloquial varieties of German).[29]

In what follows, we take a closer look at the diachronic scenario that led to the emergence of the *gehen*-middle. More precisely, we propose that the relevant grammaticalization process goes back to a conspiracy of semantic and syntactic changes, involving (at least) the following set of ingredients:

i. A semantic change in which 'to go' turns into an element expressing possibility;

ii. Analogical extension of 'go' expressing possibility to contexts in which it combines with a clausal infinitive (a control structure, with purpose clauses as a potential model/facilitating factor);

iii. Reanalysis of the control infinitive as a raising structure (necessarily linked to the availability of ambiguous bridging contexts and another

29 Historical and colloquial/dialectal varieties of German show auxiliary uses of 'go' and 'become' in a similar range of functions, including passive or future formation (see Demske 2020).

semantic change in which semi-modal 'go' loses its capability to assign a thematic role to its subject; (see Traugott 1993, Heine & Miyashita 2008 on 'threaten' and 'promise' in the history of English and German).

6.2.1 *Semantic change: development of a (im)possibility reading*

As part of the core vocabulary, the verb 'go' is typically of considerable antiquity and has developed a wide range of additional, often figurative meanings across languages. This is also true of German *gehen*, which is a massively polyfunctional/polysemic lexical element (cf. the huge entry in Grimm's German Dictionary (DWB)). In one of the potential figurative readings, 'go' expresses possibility or feasibility as e.g., in *Das geht nicht* 'That is impossible (lit. that goes not)'. Possibility-'go' may take a nominal or clausal (infinitival) argument. This is shown in (67) and (68), respectively.

(67) *In ein Gefäss von fünf Litern Inhalt gehen keine 6 1/2 Liter Wein,*
in a container of five liters content go no 6 1/2 liter wine
schon gar nicht, wenn dazu noch 3 kg Zwetschgen
even at-all not if there-to also 3 kilograms plums
und 1,8 kg Zucker darin Platz haben sollten. Das
and 1.8 kilograms sugar there-in place have should that
geht nicht, wird nie gehen, solange Zwetschgen auf Bäumen
goes not will never go as-long-as plums on trees
wachsen.
grow
'A five-litre container cannot hold six and a half litres of wine—let alone if it is also supposed to accommodate three kilograms of plums and 1.8 kilograms of sugar. That is impossible and will never be possible, as long as plums grow on trees.'
(A97/SEP.24330 St. Galler Tagblatt, 16.09.1997, Ressort: TB-LBN (Abk.); Zuviel des Guten)

(68) *Sich zurückzulehnen und darauf zu warten, was die Politik tut,*
REFL to-sit-back and there-on to wait what the politics do
geht nicht mehr.
goes not no-longer
'It is no longer possible to sit back and wait to see what politicians will do.' (NUZ17/DEZ.01471 Nürnberger Zeitung, 19.12.2017, S. 12)

This semantic development is not peculiar to German. A close parallel is Czech *jít* 'go', which Petrovičová & Škodová (2024) analyze as grammaticalizing into an analytic predicate with, among others, modal function. In its modal use, bare 'go' + infinitive is especially salient under negation (e.g. *nejde otevřít* 'it cannot be opened'), paralleling German *geht nicht zu öffnen*. The comparison is semantic rather than morphosyntactic, since Czech lacks a *zu*-analogue; the shared point is the recurrent development from motion-'go' to (im)possibility (cf. Heine & Kuteva 2002 on motion verbs as a cross-linguistically common grammaticalization source).

According to the DWB, 'go' came to be used to express possibility/feasibility from the 16th century on. Two early examples with a nominal and a clausal complement are given in (69) and (70), respectively.

(69) *Nein/ nein. Das gehet nicht.*
no no that goes not
'No, no. That is impossible.'
(DWDS; Fleming, Paul: Teütsche Poemata. Lübeck, 1642)

(70) *Doch daß man dem gerichte Die schuld aufflegen wil; geht nicht.*
however that one the.DAT court.DAT the blame on-lay want goes not
'However, it is not possible to lay the blame on the court.'
(DWDS; Gryphius, Andreas: Teutsche Reim-Gedichte. Frankfurt (Main), 1650)

The DWB mentions two potential sources of the possibility reading for 'go'. First, a related interpretation is linked to the particle verb *angehen* 'on-go', which originally expressed an inchoative meaning ('to start'), also in connection with plant growth ('to take roots, sprout'). From this, a possibility reading was derived, either with a nominal subject as in (71), or an infinitival subject clause (typically an arbitrary control structure) as in (72):

(71) *Der Käyser schüttelte den Kopff/ und sagte: Nein/ ihr redliche und Edle Nordische Helden/ das gehet nicht an!*
the emperor shook the head and said no you honest and noble Nordic heroes that goes not on
'The emperor shook his head and said: No, you honest and noble Nordic heroes, that won't do!'
(DWDS; Schupp, Johann Balthasar: Schrifften. Hrsg. v. Anton Meno Schupp. [Hanau], 1663)

(72) *den Bibliothekfonds einfach mehr zu belasten ging allerdings nicht*
the.ACC library-fund simply more to strain went however not
an, [...]
on, [...]
'However, simply putting more strain on the library fund was not an option.'
([N. N.]: Verhandlungen des Reichstages. Berlin, 1895)

Another potential source is a figurative meaning of 'go' in connection with directional adverbials, expressing that two things/workpieces potentially fit together as in (73).

(73) *das man einen eisern ring daran schieben kan, der*
that one an iron ring there-on slide can that.NOM.MASC
geheb daran gehet
exactly there-on goes
'that you can slide an iron ring on it that fits exactly'
(DWB citing Erker 8a, Frankfurt/M., 1580)

Whichever source is correct, the relevant outcome is the same: by the time the construction takes shape, 'go' has acquired stable possibility/feasibility semantics. This supplies the diachronic source of the interpretable modal feature ([iMod]) carried by raising *gehen* in Present-Day German (Section 5).

6.2.2 *Analogical extension: combination with control infinitives*

Cases such as (72) above where an instance of *angehen* 'on-go' that expresses a possibility/feasibility reading is combined with a control infinitive are robustly attested in the historical subcorpora of the DWDS.[30] What we would like to propose is that this pattern provided a model for the analogical extension of the use contexts for simple possibility-'go', which as a result came to select a control infinitive as in (74) (realized as its subject; cf. Haspelmath 1989 on the grammaticalization of the *zu*-infinitive and its compatibility with matrix predicates expressing possibility).

(74) *...um zu hören, ob es ginge, mit ihr zusammen zu reisen*
in-order to hear if it went.PRET.SBJ with her together to travel
'...to hear if it would be possible to travel with her'
(Varnhagen von Ense, Rahel: Rahel. Ein Buch des Andenkens für ihre Freunde. Bd. 2. Berlin, 1834)

Note that in (74), the clausal infinitive has undergone extraposition. In

30 A sample search query "geht nicht an #5 zu $p=VVINF" found 64 relevant hits from 1778 to 1932.

PDG, this is usually taken to be a diagnostic for a so-called incoherent construction, in which no verb cluster formation has taken place and in which the infinitival complement represents a control structure (Bech 1955/1983, Grewendorf 1988, Haider 2010).

6.2.3 *Reanalysis of control infinitives as raising structures*

The next and final step that led to the emergence of a new middle-like construction involved a reanalysis of ambiguous control infinitives as raising structures, accompanied (or facilitated) by further semantic bleaching of (then semi-modal) 'go', which lost its capability to assign a thematic role to its subject (similar to other verbs in the recorded history of the Germanic languages such as 'seem', 'threaten', or 'promise', cf. Traugott 1993, Heine & Miyashita 2008).[31]

This transition was possibly fuelled by independent changes that led to an auxiliarization of 'go' in ENHG (cf. e.g., Demske 2020). For example, other passive-like infinitival constructions, e.g. modal *sein+zu*-passives, which developed in the ENHG period (cf. Demske-Neumann 1994), may have acted as a facilitating factor, which developed in the ENHG period:

(75) *Uebers Jahr kann man sehen, ob's wird [zu schelten sein] oder [zu loben gehen].*
over-the year can one see whether=it.NOM will to reproach be or to praise.INF go.INF
'Over the course of the year, one can see whether it will turn out to deserve reproach or praise.'
(Wander, Karl Friedrich Wilhelm (Hrsg.): Deutsches Sprichwörter-Lexikon. Bd. 2. Leipzig, 1870)

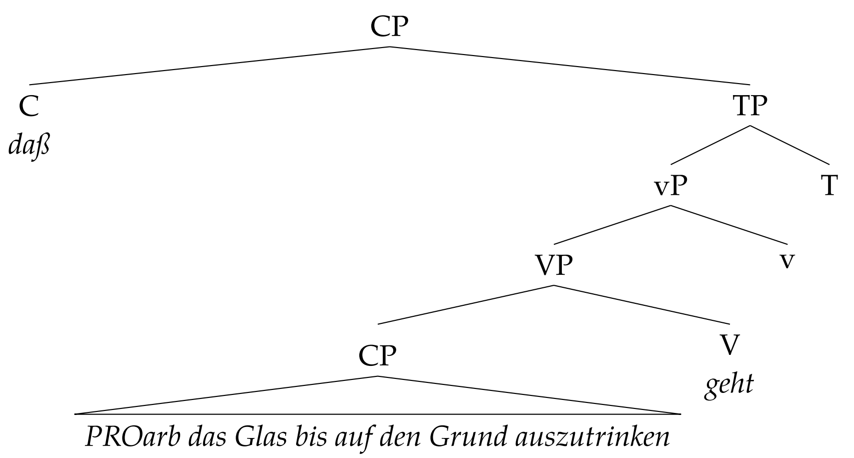
Moreover, recent work on raising and control infinitives in ENHG by Demske (2015), De Cesare (2021), and De Cesare & Demske (2025) shows that the different syntactic behaviour of raising and control infinitives (e.g., concerning extraposition of the infinitival complement) is a recent development from the eighteenth century on, lending further support to the idea that control infinitives could easily be misparsed as raising structures. Potential bridging contexts are intraposed arbitrary control infinitives in which case marking of the internal argument was not distinctive (due to common NOM/ACC syncretism in German) as in (76).

31 Control infinitives appear a likely source of raising constructions across languages.

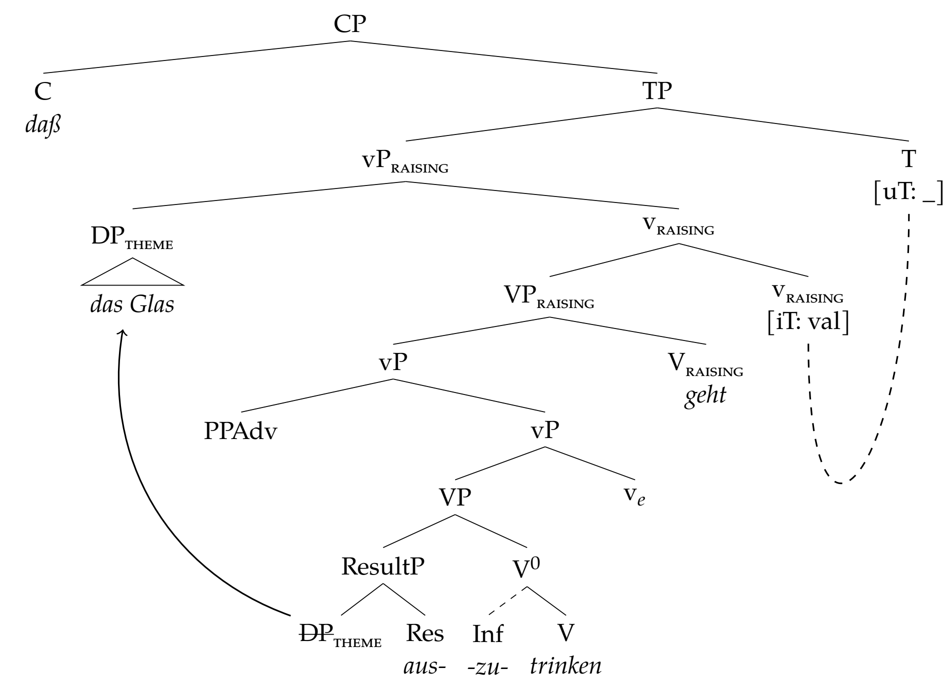
(76) *daß das Glas bis auf den Grund auszutrinken geht*
that the glass.NOM to on the bottom to-drink.INF goes
'that the glass can be drunk to the bottom'
(Pechner, Fr.: Handbuch für Lehrer beim Gebrauche des Preussischen Kinderfreundes. Der gesammte deutsche Sprachunterricht in Volksschulen oder die Uebungen im Lesen, der Grammatik, Orthographie und dem mündlichen und schriftlichen Gedankenausdrucke [...]. Königsberg, 1847)

The reanalysis of a control structure as raising infinitive leads to structural simplification in that an originally biclausal structure (involving two CPs) is reduced to a coherent verb cluster consisting of two extended verbal projections. This change is illustrated in the difference between (77) and the reanalysed structure in (78). In (78) the adverbial *bis auf den Grund*, present in (77), is omitted for ease of exhibition.

(77)



(78)



The loss of a fully biclausal structure brings the core changes needed for the synchronic pattern. 'Go' becomes a raising verb: it no longer assigns a thematic role to its subject, and it selects a reduced verbal complement rather than a CP capable of licensing PRO. The former clausal subject of the matrix predicate is reanalysed as the complement of raising *gehen*; in this reduced, Voice-less *v*P, no active *v* remains to assign accusative, so the internal argument raises to matrix TP for nominative case. The result is precisely the voiceless *v*P selected by *gehen* in the synchronic analysis (Section 5).

As a result of this reanalysis, genuine examples of the *gehen*-middle should lack extraposition of the infinitival complement, since raising (and other clause union) constructions require verb clustering in German (cf. Haider 2010). This is in line with our observation that the *zu*-infinitive always directly precedes finite 'go' in all cases, where the *gehen*-middle is part of an embedded clause.[32] Moreover, we expect new raising patterns to appear in which *gehen* clearly combines with a DP subject. Relevant diagnostics include (i) unambiguous nominative marking on the subject DP and (ii) plural agreement on *gehen*. Some relevant examples are given in (79) and (80).

32 But note that the loss of extraposition in raising structures was probably an independent, slightly earlier or parallel development (cf. Demske 2015, De Cesare 2021, De Cesare & Demske 2025).

(79) ...*diesen entschiedenen Gegensatz, welcher sichtbar und empfindbar*
...this clear opposition which visible and perceptible
ist und ***der*** *nicht aufzuheben geht*
is and that.MASC.NOM not eliminated.INF goes

'this clear opposition, which is visible and perceptible, and which cannot be eliminated'
(Johann Wolfgang Goethe, Zur Farbenlehre, 1810)

(80) *Da diese letzteren am leichtesten abzuändern* ***gehen*** [...]
since the latter.PL on-the easiest modify.INF go.PL

'Since the latter are the easiest to modify...'
(Theodor Pregél, Die Ermittelung der Stufenscheibenverhältnisse für Werkzeugmaschinen. In: Dinglers Polytechnisches Journal, 1893/287, pp. 248-252. Stuttgart, 1893)

Hence, we should expect an increase in cases that clearly involve raising (relative to the number of cases that are ambiguous between raising and control). This expectation is, however, only borne out for the short period in the late nineteenth century, where the use of the *gehen*-middle was apparently more common (see Figure 12). The absence of a clear-cut pattern may also be attributable to the limited size of the data set (n=95). Regardless, additional data from the eighteenth to the early twentieth century is necessary for a more robust understanding of the phenomenon.

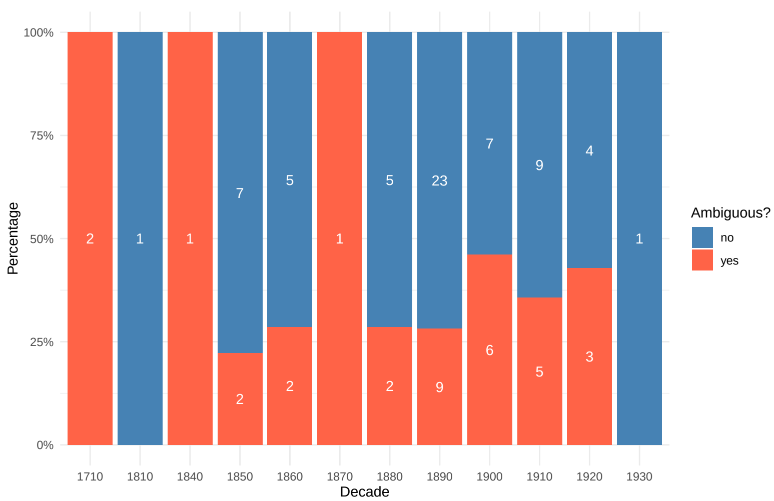


Figure 12 Proportion of ambiguous occurrences over time (per decade with absolute numbers)

7 Concluding summary

The first part of this paper presented a novel in-depth investigation of [*gehen*+*zu*-INFINITIVE]. Using experimental methodology and limited qualitative *post-hoc* syntactic and semantic testing, we were able to confidently classify [*gehen*+*zu*-INFINITIVE] as an infinitival construction expressing middle voice. It is neither an infinitival passive (*contra* Holl 2010, Abraham 2020, Müller 2019) nor a periphrastic (impossibility) anticausative (*contra* Cysouw 2023). The behaviour of this middle is determined by constraints involving argument structure, aspect, and modal scope. First, *gehen* is a semi-modal (possibility) unaccusative *v*-raising head, but lacks any eventive or thematic properties itself. Second, this head only selects an eventive *v*P, ruling out Agent introduction in a lower VoiceP. Aspectual restrictions are instead enforced at C/I and LF. As a modal operator, *gehen* requires a structurally distinct target state to evaluate. We argued that only causative *v*P-configurations can provide this environment, via merge of a complement ResultP. Together, these factors account for the passive-like interpretation, the dispositional properties of the promoted argument, and the lexical-semantic restriction to Accomplishments. Conversely, the semantic and syntactic properties associated with Achievements and Activities fail to project an aspectually compatible result state, causing the derivation to fail at the interfaces.

The second part of the paper dealt with the historical development of the *gehen*-middle. Based on a study in the historical subcorpora of the DWDS corpus, the construction presumably developed in the early eighteenth century via a reanalysis of ambiguous control structures as raising infinitives, which gave rise to a (semi-modal) raising version of 'go'. This structural shift was made possible by an earlier semantic change in which 'go' came to express possibility/feasibility. However, we have also seen that the heyday of the new pattern was rather short-lived, giving rise to the impression of a failed change. Possibly, this has to do with the fact that the *gehen*-middle has never lost its colloquial character (which initially might have contributed to its attractiveness), which ultimately led to its demise in the written language.

A Appendix: Raw Data

Data for: Compound Tense

Condition	**Item (Sentence)**	*M*	*SD*	M_z	SD_z
scheinen_past	Die Journalistin schien gut informiert zu sein.	4.92	0.33	1.39	0.41
gehen_past	Der Wagen ging nicht zu reparieren.	3.51	1.39	0.42	0.69
gehen_past	Der Wagen ist nicht zu reparieren gegangen.	1.44	0.77	-0.95	0.52
scheinen_past	Die Journalistin hat gut informiert zu sein geschienen.	1.47	0.87	-0.94	0.62

Table 4 Descriptive Statistics: Compound Tense

Data for: Extraposition

Condition	**Item (Sentence)**	*M*	*SD*	M_z	SD_z
gehen_!extraposition	Louisa ist sehr glücklich, dass das Spiel endlich zu installieren geht .	3.31	1.15	0.30	0.56
gehen_extraposition	Louisa ist sehr glücklich, dass das Spiel endlich geht zu installieren.	1.31	0.62	-1.03	0.44

Table 5 Descriptive Statistics: Extraposition

Data for: Datives and Double Arguments

Condition	**Item (Sentence)**	*M*	*SD*	M_z	SD_z
sein_dat	Ihm ist nicht zu helfen.	4.81	0.64	1.32	0.56
gehen+distransitive	Die Email geht ihm nicht zu schicken.	1.17	0.50	-1.14	0.37
gehen_dat	Ihm geht nicht zu helfen.	1.38	0.68	-0.99	0.45

Table 6 Descriptive Statistics: Datives and Double Arguments

Data for: Agentive von-phrases and für-phrases

Condition	Item (Sentence)	*M*	*SD*	M_z	SD_z
werden+von	Das Schmetterlingshaus wird von allen Besuchern geliebt.	4.78	0.51	1.30	0.46
sind+für	Die Fresken sind jetzt wieder für alle Besucher zu sehen.	4.69	0.52	1.23	0.40
lassen+von	Der Bürgermeister lässt eine Mauer von den Bauarbeitern errichten.	4.32	0.95	0.97	0.54
gehören+von	Dieser laute Krach gehört von der Polizei verboten.	3.89	1.11	0.68	0.67
middle+für	Das Buch liest sich für sie gut.	3.65	1.24	0.54	0.77
gehen+für	Der Tisch ging für den Lehrer einfach abzuwischen, aber nicht für die Kinder.	2.92	1.36	0.03	0.73
sind+von	Die Fresken sind jetzt wieder von allen Besuchern zu sehen.	2.88	1.27	0.03	0.82
gehen+für	Die Tür geht für Anna kaum aufzuschließen.	2.76	1.32	-0.06	0.68
gehen+für	Der Blinker geht für Younes leicht zu reparieren.	2.58	1.22	-0.19	0.59
gehen+von	Der Tisch ging vom Lehrer einfach abzuwischen, aber nicht von den Kindern.	2.12	1.19	-0.49	0.63
gehen+von	Die Tür geht von Anna kaum aufzuschließen.	1.81	1.03	-0.69	0.57
gehen+von	Der Blinker geht von Younes leicht zu reparieren.	1.62	0.83	-0.83	0.43
middle+von	Das Buch liest sich von ihr gut.	1.47	0.87	-0.92	0.58

Table 7 Descriptive Statistics: Agentive von-phrases and für-phrases

Data for: Instrumental adjuncts

Condition	Item (Sentence)	*M*	*SD*	M_z	SD_z
gehen+instrument	Die Schraube geht aber mit einem besonderen Schraubenzieher leicht zu lösen.	3.21	1.35	0.21	0.75

Table 8 Descriptive Statistics: Instrumental adjuncts

Data for: Adverbial Modification

Condition	Item (Sentence)	*M*	*SD*	M_z	SD_z
gehen+otheradv	Die Tür geht kaum aufzuschließen.	3.85	1.27	0.64	0.70
gehen+midadv	Die Tür geht jetzt leicht aufzuschließen.	3.39	1.20	0.35	0.58
gehen+midadv	Der Tisch ging dann ganz einfach abzuwischen.	3.46	1.36	0.39	0.70
gehen+otheradv	Die Tür geht jetzt wieder aufzuschließen.	3.43	1.45	0.37	0.74
gehen+nicht	Die Schraube geht nicht einfach zu lösen.	3.25	1.33	0.25	0.69
gehen+nicht	Der gefrorene Apfel ging nicht zu essen.	3.08	1.32	0.14	0.48
gehen+otheradv	Die Tür geht jetzt aufzuschließen.	2.99	1.38	0.07	0.72
gehen+otheradv	Der Tisch ging zuerst nicht abzuwischen.	2.71	1.31	-0.10	0.66
gehen+otheradv	Der Quark geht noch zu essen.	2.25	1.20	-0.40	0.66

Table 9 Descriptive Statistics: Adverbial Modification

Data for Vendler's Verb Classes

Class	Condition	Item (Sentence)	*M*	*SD*	M_z	SD_z
Accomplishment	gehen+midadv	DieTür geht jetzt leicht aufzuschließen.	3.76	1.24	0.58	0.63
Accomplishment	gehen_past	Der Wagen ging nicht zu reparieren.	3.51	1.39	0.42	0.69
Accomplishment	gehen+midadv	Der Tisch ging dann ganz einfach abzuwischen.	3.46	1.37	0.39	0.70
Accomplishment	gehen+nicht	Die Schraube geht nicht einfach zu lösen.	3.25	1.34	0.25	0.69
Accomplishment	gehen+nicht	Das alte Spiel geht nicht zu installieren.	3.21	1.30	0.22	0.67
Accomplishment	gehen+nicht	Das geht leider nicht zu ändern.	2.94	1.35	0.06	0.76
Achievement	gehen+nicht	Der Schaffner geht nicht zu überzeugen.	1.75	0.90	-0.72	0.49
Achievement	gehen+nicht	Der Berggipfel geht einfach nicht zu erreichen.	1.78	0.86	-0.71	0.46
Achievement	gehen+nicht	Robert geht einfach nicht zu schlagen.	1.81	0.99	-0.68	0.51
Achievement	gehen+nicht	Superman geht nicht zutöten.	2.12	1.20	-0.49	0.63
Achievement	gehen+nicht	Die Schwierigkeiten gingen nicht zu überwinden.	2.17	1.06	-0.45	0.55
Activity	gehen+adv	Jimmys Gitarre geht leicht zu spielen.	3.39	1.34	0.34	0.74
Activity	gehen+otheradv	Der Quark geht noch zu essen.	2.25	1.21	-0.40	0.66
Activity	gehen+nicht	Der gefrorene Apfel ging nicht zu essen.	2.61	1.19	-0.17	0.61

Table 10 Descriptive Statistics: Vendler's Verb Classes

B Appendix: Experimental Stimuli and Descriptive Statistics

Table 11: Descriptive Statistics for All Experimental Items

Condition	Item (Original & Intended Translation)	*M*	*SD*	M_z	SD_z
1sg	Ich gehe nicht zu ändern? Warum lässt Carla mich nicht einfach so wie ich bin?! *'I can't be changed? Why doesn't Carla just leave me the way I am?!'*	1.56	0.75	-0.87	0.41
2sg	Du gehst einfach nicht zu ändern. *'You simply cannot be changed.'*	1.26	0.67	-1.07	0.46
Probe_gut	Der Käsekuchen wurde von meinem Bruder aufgegessen. *'The cheesecake was eaten up by my brother.'*	4.68	0.62	1.21	0.46
Probe_schlecht	Von meinem Brüdern wurde aufgegesst den Käsekuchen. *'By my brothers was eaten up the cheesecake. (ungrammatical)'*	1.03	0.17	-1.25	0.44
gehen_!extraposition	Louisa ist sehr glücklich, dass das Spiel endlich zu installieren geht. *'Louisa is very happy that the game can finally be installed. (intraposed)'*	3.29	1.40	0.30	0.74
gehen_!zu	Der Wagen ging nicht reparieren. *'The car could not be repaired. (missing zu)'*	1.15	0.49	-1.13	0.44

Continued on next page

Table 11 – continued from previous page

Condition	**Item (Original & Translation)**	M	SD	M_z	SD_z
gehen_dat	Ihm geht nicht zu helfen. *'He cannot be helped. (gehen)'*	1.38	0.68	-0.99	0.41
gehen_extraposition	Louisa ist sehr glücklich, dass das Spiel endlich geht zu installieren. *'Louisa is very happy that the game can finally be installed. (extraposed)'*	1.29	0.76	-1.03	0.52
gehen_past	Der Wagen ging nicht zu reparieren. *'The car could not be repaired.'*	3.51	1.39	0.42	0.72
gehen_past	Der Wagen ist nicht zu reparieren gegangen. *'The car has not been able to be repaired. (compound)'*	1.43	0.73	-0.95	0.45
gehen+adv	Jimmys Gitarre geht leicht zu spielen. *'Jimmy's guitar can easily be played.'*	3.39	1.18	0.34	0.56
gehen+distransitive	Die Email geht ihm nicht zu schicken. *'The email cannot be sent to him.'*	1.17	0.50	-1.14	0.41
gehen+für	Die Tür geht für Anna kaum aufzuschließen. *'The door can barely be unlocked for Anna.'*	2.76	1.32	-0.06	0.68
gehen+für	Der Blinker geht für Younes leicht zu reparieren. *'The indicator can easily be repaired for Younes.'*	2.75	1.32	-0.09	0.73
gehen+für	Der Tisch ging für den Lehrer einfach abzuwischen, aber nicht für die Kinder. *'The table could simply be wiped off for the teacher, but not for the children.'*	2.74	1.25	-0.10	0.61
gehen+instrument	Die Schraube geht aber mit einem besonderen Schraubenzieher leicht zu lösen. *'The screw can however easily be loosened with a special screwdriver.'*	3.21	1.35	0.21	0.75
gehen+midadv	Der Tisch ging dann ganz einfach abzuwischen. *'The table could then be wiped off quite simply.'*	3.46	1.36	0.39	0.70
gehen+midadv	Die Tür geht jetzt leicht aufzuschließen. *'The door can easily be unlocked now.'*	3.39	1.20	0.35	0.58

Continued on next page

Table 11 – continued from previous page

Condition	**Item (Original & Translation)**	M	SD	M_z	SD_z
gehen+nicht	Das alte Spiel geht nicht zu installieren. *'The old game cannot be installed.'*	3.65	1.46	0.49	0.75
gehen+nicht	Das geht leider nicht zu ändern. *'Unfortunately, that cannot be changed.'*	3.57	1.36	0.44	0.69
gehen+nicht	Die Schraube geht nicht einfach zu lösen. *'The screw cannot simply be loosened.'*	3.25	1.33	0.25	0.69
gehen+nicht	Der gefrorene Apfel ging nicht zu essen. *'The frozen apple could not be eaten.'*	3.08	1.32	0.14	0.48
gehen+nicht	Der Schaffner geht nicht zu überzeugen. *'The conductor cannot be convinced.'*	1.94	1.09	-0.58	0.66
gehen+nicht	Die Schwierigkeiten gingen nicht zu überwinden. *'The difficulties could not be overcome.'*	1.83	0.95	-0.67	0.56
gehen+nicht	Der Berggipfel geht einfach nicht zu erreichen. *'The mountain peak simply cannot be reached.'*	1.79	0.99	-0.69	0.55
gehen+nicht	Robert geht einfach nicht zu schlagen. *'Robert simply cannot be beaten.'*	1.62	0.88	-0.79	0.53
gehen+nicht	Superman geht nicht zu töten. *'Superman cannot be killed.'*	1.47	0.77	-0.89	0.50
gehen+otheradv	Die Tür geht kaum aufzuschließen. *'The door can barely be unlocked.'*	3.85	1.27	0.64	0.70
gehen+otheradv	Die Tür geht jetzt wieder aufzuschließen. *'The door can be unlocked again now.'*	3.43	1.45	0.37	0.74
gehen+otheradv	Die Tür geht jetzt aufzuschließen. *'The door can be unlocked now.'*	2.99	1.38	0.07	0.72
gehen+otheradv	Der Tisch ging zuerst nicht abzuwischen. *'The table could not be wiped off at first.'*	2.71	1.31	-0.10	0.66
gehen+otheradv	Der Quark geht noch zu essen. *'The quark can still be eaten.'*	2.25	1.20	-0.40	0.66

Continued on next page

Table 11 – continued from previous page

Condition	Item (Original & Translation)	M	SD	M_z	SD_z
gehen+von	Der Tisch ging vom Lehrer einfach abzuwischen, aber nicht von den Kindern. *'The table could simply be wiped off by the teacher, but not by the children.'*	2.12	1.19	-0.49	0.63
gehen+von	Die Tür geht von Anna kaum aufzuschließen. *'The door can barely be unlocked by Anna.'*	1.81	1.03	-0.69	0.57
gehen+von	Der Blinker geht von Younes leicht zu reparieren. *'The indicator can easily be repaired by Younes.'*	1.64	0.86	-0.81	0.46
gehört+von	Dieser laute Krach gehört von der Polizei verboten. *'This loud noise ought to be forbidden by the police.'*	3.76	1.25	0.60	0.64
gehört-von	Dieser laute Krach gehört verboten. *'This loud noise ought to be forbidden.'*	4.01	1.27	0.76	0.70
lassen_DO	Der Bürgermeister lässt die Bauarbeiter eine Mauer errichten. *'The mayor lets the construction workers build a wall.'*	4.67	0.53	1.19	0.35
lassen+von	Der Bürgermeister lässt eine Mauer von den Bauarbeitern errichten. *'The mayor lets a wall be built by the construction workers.'*	4.31	0.86	0.97	0.54
middle	Das Buch liest sich gut. *'The book reads well.'*	4.54	0.69	1.11	0.45
middle+für	Das Buch liest sich für sie gut. *'The book reads well for her.'*	3.65	1.24	0.54	0.77
middle+von	Das Buch liest sich von ihr gut. *'The book reads well by her.'*	1.47	0.87	-0.92	0.58
scheinen_past	Die Journalistin schien gut informiert zu sein. *'The journalist seemed to be well informed.'*	4.79	0.41	1.30	0.23
scheinen_past	Die Journalistin hat gut informiert zu sein geschienen. *'The journalist has seemed to be well informed. (compound)'*	1.39	0.62	-0.94	0.38

Continued on next page

Table 11 – continued from previous page

Condition	Item (Original & Translation)	M	SD	M_z	SD_z
sein_dat	Ihm ist nicht zu helfen. *'He cannot be helped. (sein)'*	4.81	0.64	1.32	0.37
sind+für	Die Fresken sind jetzt wieder für alle Besucher zu sehen. *'The frescoes can be seen again now for all visitors.'*	4.69	0.52	1.23	0.40
sind+von	Die Fresken sind jetzt wieder von allen Besuchern zu sehen. *'The frescoes can be seen again now by all visitors.'*	2.88	1.27	0.03	0.82
werden+von	Das Schmetterlingshaus wird von allen Besuchern geliebt. *'The butterfly house is loved by all visitors.'*	4.75	0.55	1.25	0.46
zu_inf+pl_werden	obwohl die Brücken zu reparieren versucht wurden. *'although the bridges were attempted to be repaired. (PL)'*	2.39	1.17	-0.28	0.70
zu_inf+sg_werden	obwohl die Brücken zu reparieren versucht wurde. *'although it was attempted to repair the bridges. (SG)'*	1.64	0.83	-0.83	0.43

References

Abraham, Werner. 1993. Null subjects in the history of German: From IP to CP. *Lingua* 89(2). 117–142. doi:https://doi.org/10.1016/0024-3841(93)90050-7.

Abraham, Werner. 2020. *Modality in syntax, semantics and pragmatics*, vol. 165. Cambridge, UK: Cambridge University Press.

Ackema, Peter & Ad Neeleman. 2004. *Beyond morphology: Interface conditions on word formation*. Oxford: Oxford University Press.

Ackema, Peter & Maaike Schoorlemmer. 1994. The middle construction and the syntax-semantics interface. *Lingua* 93(1). 59–90. doi:10.1016/0024-3841(94)90353-0.

Ackema, Peter & Maaike Schoorlemmer. 1995. Middles and Non-Movement. *Linguistic Inquiry* 26. 173–197.

Ackema, Peter & Maaike Schoorlemmer. 2015. Middles. In *The Blackwell companion to syntax*, Oxford, UK: Blackwell 2nd edn.

Adger, David. 2003. *Core syntax: A Minimalist approach*. Oxford University Press.

Alexiadou, Artemis, Elena Anagnostopoulou & Florian Schäfer. 2006a. The fine structure of (anti-)causatives. In Chris Davis, Amy Rose Deal & Youri Zabbal (eds.), *Proceedings of NELS 36*, 115–128. Amherst, MA: GLSA.

Alexiadou, Artemis, Elena Anagnostopoulou & Florian Schäfer. 2006b. The properties of anticausatives crosslingustically. In Mara Frascarelli (ed.), *Phases of Interpretation*, 187–211. Berlin: Mouton de Gruyter.

Alexiadou, Artemis, Elena Anagnostopoulou & Florian Schäfer. 2015. *External arguments in transitivity alternations: A layering approach*. Oxford: Oxford University Press.

Alexiadou, Artemis, Elena Anagnostopoulou & Florian Schäfer. 2018. Passive. In Norbert Hornstein, Howard Lasnik, Pritty Patel-Grosz & Charles Yang (eds.), *Syntactic Structures after 60 years: The impact of the Chomskyan revolution in linguistics*, 403–425. Mouton de Gruyter.

Alexiadou, Artemis & Florian Schäfer. 2011. *There*-insertion: An unaccusativity mismatch at the syntax-semantics interface. In Mary Byram Washburn, Sarah Ouwayda, Chuoying Ouyang, Bin Yin, Canan Ipek, Lisa Marston & Aaron Walker (eds.), *Online proceedings of WCCFL 28*, Los Angeles, CA: University of Southern California.

Arad, Maya. 2003. Locality constraints on the interpretation of roots: The case of Hebrew denominal verbs. *Natural Language and Linguistic Theory* 21. 737–778.

Baker, Mark, Kyle Johnson & Ian Roberts. 1989. Passive arguments raised. *Linguistic Inquiry* 20(2). 219–251.

Beavers, John. 2008. Scalar complexity and the structure of events. In Johannes Dölling, Tatjana Heyde-Zybatow & Martin Schäfer (eds.), *Event structures in linguistic form and interpretation*, 245–266. Berlin, Boston: De Gruyter. doi:10.1515/9783110925449.245.

Bech, Gunnar. 1955/1983. *Studien über das deutsche verbum infinitum* (Linguistische Arbeiten 139). Tübingen: Max Niemeyer Verlag 2nd edn. doi:none. Zuerst erschienen 1955.

Behaghel, Otto. 1923. *Deutsche Syntax: eine geschichtliche Darstellung; Bd.ii Die Wortklassen und Wortformen: A: Nomen, Pronomen*. Heidelberg: Winter.

Bhatt, Rajesh. 2006. *Covert modality in non-finite contexts*, vol. 8 Interface Explorations [IE]. Berlin: Mouton de Gruyter.

Bhatt, Rajesh & Roumyana Pancheva. 2006. Implicit Arguments. In *The Blackwell Companion to Syntax*, vol. 2, Blackwell.

Biberauer, Theresa & Ian Roberts. 2010. Subjects, tense and verb-movement. In Theresa Biberauer, Anders Holmberg, Ian Roberts & Michelle Sheehan (eds.), *Parametric variation: null subjects in minimalist theory*, 263–303. Cambridge: Cambridge University Press Cambridge.

Biberauer, Theresa & Ian Roberts. 2017. Parameter setting. In Adam Ledgeway & Ian Roberts (eds.), *The cambridge handbook of historical syntax*, 134–162. Cambridge University Press. doi:10.1017/9781107279070.008.

Bjorkman, Bronwyn. 2011. *BE-ing default: The morphosyntax of auxiliaries*: Massachusetts Institute of Technology dissertation.

Bobaljik, Jonathan & Susi Wurmbrand. 2005. The domain of agreement. *Natural Language and Linguistic Theory* 23(4). 809–865. doi:10.1007/s11049-004-3792-4.

Boon, Pieter. 1981. Der Gebrauch des sog. „modalen Infinitivs" in den Verbgefügen *sein* + Infinitiv mit *zu* bzw. *haben* + Infinitiv mit *zu* durch Thomas Murner. *Beiträge zur Erforschung der deutschen Sprache* (1). 191–198.

Borer, Hagit. 2005. *Structuring sense, vol. 2: The normal course of events*. Oxford: Oxford University Press.

Brinkmann, Hennig. 1962. *Die deutsche Sprache: Gestalt und Leistung*. Düsseldorf: Schwann.

Broekhuis, Hans. 2023. VO or OV: V to *v* or not to *v*. *Linguistic Variation* 23(2). 343–378. doi:10.1075/lv.22005.bro.

Bruening, Benjamin. 2013. *By* phrases in passives and nominals. *Syntax* 16. 1–41.

Chierchia, Gennaro. 2004. A semantics for unaccusatives and its syntactic consequences. In Artemis Alexiadou, Elena Anagnostopoulou & Martin Everaert (eds.), *The unaccusativity puzzle: Explorations of the syntax-lexicon interface*, 22–59. Oxford: Oxford University Press.

Chomsky, Noam. 1981. *Lectures on Government and Binding*. Dordrecht: Foris.

Chomsky, Noam. 1995. *The minimalist program*. Cambridge, MA: MIT Press.

Chomsky, Noam. 2000. Minimalist inquiries: the framework. In Roger Martin, David Michaels & Juan Uriagereka (eds.), *Step by step: Essays on Minimalist syntax in honor of Howard Lasnik*, 89–155. Cambridge, MA: MIT Press.

Chomsky, Noam. 2001. Derivation by phase. In Michael Kenstowicz (ed.), *Ken Hale: A life in language*, 1–52. Cambridge, MA: MIT Press.

Chomsky, Noam. 2008. On Phases. In Robert Freidin, Carlos P. Otero & Maria Luisa Zubizarreta (eds.), *Foundational issues in linguistic theory. essays in honor of Jean-Roger Vergnaud*, 133–166. Cambridge, MA: Cambridge, MA.

Cinque, Guglielmo. 1999. *Adverbs and Functional Heads: A Cross-Linguistic Perspective* Oxford studies in Comparative Syntax. New York: Oxford University Press.

Cinque, Guglielmo. 2006a. *Restructuring and functional heads*. Oxford: Oxford University Press.

Cinque, Guglielmo. 2006b. *Restructuring and Functional Heads* Oxford studies in Comparative Syntax. New York: Oxford University Press.

Condoravdi, Cleo. 1989. The middle: where semantics and morphology meet. *MIT Working Papers in Linguistics* 11. 16–30.

Cornips, Leonie & Cecilia Poletto. 2005. On collecting dialect data: The role of word order, subject-verb agreement, and relative clauses. *Lingua* 115(7). 939–957.

Cysouw, Michael. 2023. *Encyclopaedia of German diatheses* (Open Germanic Linguistics 4). Berlin: Language Science Press. doi:10.5281/zenodo.7602514.

Davies, William D & Stanley Dubinsky. 2004. *The grammar of raising and control: A course in syntactic argumentation*. Malden, MA: Blackwell Pub. doi:10.1002/9780470755693.

De Cesare, Ilaria. 2021. *Word order variability and change in German infinitival complements. A multi-causal approach*: University of Potsdam Doctoral dissertation.

De Cesare, Ilaria & Ulrike Demske. 2025. Aspectual verbs in German: A diachronic view. *Journal of Germanic Linguistics* 2(37). 163–189.

Demske, Ulrike. 2015. Towards coherent infinitival patterns in the history of German. *Journal of Historical Linguistics* 5(1). 6–40. doi:10.1075/jhl.5.1.01dem. https://doi.org/10.1075/jhl.5.1.01dem.

Demske, Ulrike. 2020. Zur Grammatikalisierung von 'gehen' im Deutschen. In *Forum japanisch-germanistischer sprachforschung*, vol. 2, 9–42.

Demske-Neumann, Ulrike. 1994. *Modales Passiv und Tough Movement. Zur strukturellen Kausalität eines syntaktischen Wandels im Deutschen und Englischen*. Tübingen: Niemeyer.

den Dikken, Marcel. 2007. Phase extension: Contours of a theory of the role of head movement in phrasal extraction. *Theoretical Linguistics* 33. 1–41.

Dobler, Eva. 2008. Again'and the structure of result states. *Proceedings of ConSOLE XV* 1. 42–66.

Dowty, David. 1979. *Word meaning and montague grammar*. Dordrecht: Reidel.

Ebert, Robert Peter. 1976. *Infinitival complement constructions in Early New High German*, vol. 30 LA. Berlin, New York: De Gruyter. doi:10.1515/9783111355832.

Embick, David. 2004a. On the structure of resultative participles in English. *Linguistic Inquiry* 35(3). 355–392.

Embick, David. 2004b. Unaccusative syntax and verbal alternations. In Artemis Alexiadou, Elena Anagnostopoulou & Martin Everaert (eds.), *The unaccusativity puzzle: Explorations of the syntax-lexicon interface*, 137–158. Oxford: Oxford University Press.

Embick, David. 2010. *Localism versus globalism in morphology and phonology* (Linguistic Inquiry Monographs 60). Cambridge, MA: MIT Press.

Embick, David. 2015. *The Morpheme: A Theoretical Introduction* Interface Explorations. Berlin, Boston: De Gruyter Mouton.

Embick, David & Rolf Noyer. 2001. Movement operations after syntax. *Linguistic Inquiry* 32(4). 555–595.

Eroms, Hans-Werner. 1990. Zur Entwicklung der Passivperiphrasen im Deutschen. In Anne Betten & Claudia Maria Riehl (eds.), *Neuere Forschungen zur historischen Syntax des Deutschen*, 82–97. Berlin: De Gruyter. doi:10.1515/9783111708751.82.

Fagan, Sarah. 1992. *The syntax and semantics of middle constructions: A study with special reference to German*. Cambridge, UK: Cambridge University Press.

Featherston, Sam. 2005. Magnitude estimation and what it can do for your syntax. *Lingua* 115(11). 1525–1550.

Featherston, Sam. 2007. Data in Generative Grammar: The stick and the carrot. *Theoretical Linguistics* 33(3). 269–318.

Folli, Raffaella & Heidi Harley. 2005. Flavors of v. In Paula Kempchinsky & Roumyana Slabakova (eds.), *Aspectual inquiries*, 95–120. Dordrecht: Springer.

Folli, Raffaella & Heidi Harley. 2007. Causation, obligation, and argument structure: On the nature of little v. *Linguistic Inquiry* 38(2). 197–218. doi:10.1162/ling.2007.38.2.197.

Fuß, Eric. 2008. *Word order and language change: On the interface between syntax and morphology*: University of Frankfurt/M. Habilitation.

Gaeta, Livio. 2018. Im Passiv sprechen in den Alpen. *Sprachwissenschaft* 2(43). 221–250.

Grewendorf, Günther. 2001. Multiple wh-fronting. *Linguistic Inquiry* 32(1). 87–122. doi:10.1162/002438901554595.

Grewendorf, Günther & Joachim Sabel. 1999. Scrambling in German and Japanese: Adjunction versus multiple specifiers. *Natural Language & Linguistic Theory* 17(1). 1–65. doi:10.1023/A:1006068326583.

Grewendorf, Günther. 1988. *Aspekte der deutschen Syntax: Eine Rektions-Bindungs-Analyse* (Studien zur deutschen Grammatik 33). Tübingen: Gunter Narr Verlag.

Grewendorf, Günther. 1989. *Ergativity in german*, vol. 35 SGG. Berlin, Boston: De Gruyter. doi:10.1515/9783110859256.

Haiden, Martin. 2005. *Theta theory*, vol. 78. Berlin, New York: Mouton de Gruyter. doi:10.1515/9783110197471.

Haider, Hubert. 1993. *Deutsche Syntax, generativ: Vorstudien zur Theorie einer projektiven Grammatik*. Tübingen: Narr.

Haider, Hubert. 1997. Typological implications of a directionality constraint on projections. In Artemis Alexiadou & Tracy Hall (eds.), *Studies on universal grammar and typological variation*, 17–33. Amsterdam: John Benjamins.

Haider, Hubert. 2005. How to turn German into Icelandic — and derive the OV–VO contrasts. *The Journal of Comparative Germanic Linguistics* 8(1). 1–56. doi:10.1007/s10828-004-0293-0.

Haider, Hubert. 2010. *The syntax of German*. Cambridge: Cambridge University Press.

Hale, Ken & Samuel Jay Keyser. 1993. On argument structure and the lexical expression of syntactic relations. In Ken Hale & Samuel Jay Keyser (eds.), *The view from building 20. essays in linguistics in honor of Sylvain Bromberger*, Cambridge, MA: MIT Press.

Hale, Ken & Samuel Jay Keyser. 2002. *Prolegomenon to a theory of argument structure* (Linguistic Inquiry Monographs 39). Cambridge, MA: MIT Press.

Hale, Kenneth & Samuel J. Keyser. 1987. *A view from the middle* (Lexicon Project Working Papers 10). Cambridge, MA: MIT Working Papers in Linguistics.

Halle, Morris. 1997. Distributed Morphology: Impoverishment and Fission. In Benjamin Bruening, Yoonjung Kang & Martha McGinnis (eds.), *MIT working papers in linguistics 30: PF: Papers at the interface*, 425–449. MIT Press.

Halle, Morris & Alec Marantz. 1993. Distributed Morphology and the pieces of inflection. In Ken Hale & Samuel Jay Keyser (eds.), *The view from building 20: Essays in linguistics in honor of Sylvain Bromberger*, 111–176. Cambridge, MA: MIT Press.

Harley, Heidi. 2005. How do verbs get their names? denominal verbs, manner incorporation, and the ontology of verb roots in English. In Nomi Ertschik-Shir & Tova Rapoport (eds.), *The syntax of aspect*, 42–64. Oxford: Oxford University Press.

Haspelmath, Martin. 1989. From purposive to infinitive - a universal pathway of grammaticalization. *Folia Linguistica Historica* 23(X/1-2). 287–310. doi:10.1515/flih.1989.10.1-2.287.

Haspelmath, Martin. 1990. The grammaticalization of passive morphology. *Studies in Language* 14(1). 25–72. doi:10.1075/sl.14.1.03has.

Hein, Johannes. 2021. Verb movement and the lack of verb-doubling VP-topicalization in Germanic. *The Journal of Comparative Germanic Linguistics* 24(1). 89–144. doi:10.1007/s10828-021-09125-5.

Heine, Bernd & Tania Kuteva. 2002. *World lexicon of grammaticalization*. Cambridge: Cambridge University Press 1st edn.

doi:10.1017/CBO9780511613463.

Heine, Bernd & Hiroyuki Miyashita. 2008. Accounting for a functional category: German drohen 'to threaten'. *Language Sciences* 1(30). 53–101.

Higginbotham, James. 2000. Accomplishments. In *Proceedings of glow in asia ii*, vol. 2, 72–82. Nagoya: Nanzan University.

Höhle, Tilman N. 1978. *Lexikalistische syntax. die aktiv-passiv-relation und andere infinitkonstruktionen im deutschen*, vol. 67 Linguistische Arbeiten. Berlin, New York: De Gruyter. doi:10.1515/9783111345444.

Holl, Daniel. 2010. *Modale Infinitive und dispositionelle Modalität im Deutschen*. Berlin: Akademie Verlag. doi:10.1524/9783050062341.

Höhle, Tilman N. 1978. *Lexikalistische Syntax. Die Aktiv-Passiv-Relation und andere Infinitkonstruktionen im Deutschen*, vol. 67 LA. Berlin, New York: De Gruyter. doi:10.1515/9783111345444. http://www.degruyter.com/view/books/9783111345444/9783111345444/9783111345444.xml.

Keyser, Samuel J. & Thomas Roeper. 1984. On the Middle and Ergative Constructions in English. *Linguistic Inquiry* 15. 381–416.

Kiparsky, Paul. 1973. "Elsewhere" in phonology. In Stephen R. Anderson & Paul Kiparsky (eds.), *A festschrift for morris halle*, 93–106. Holt, Rinehart, and Winston.

Kratzer, Angelika. 1991. Modality. In Arnim von Stechow & Dieter Wunderlich (eds.), *Semantics: An international handbook of contemporary research*, 639–650. Berlin: de Gruyter.

Kratzer, Angelika. 1996. Severing the external argument from its verb. In Johan Rooryck & Laurie Ann Zaring (eds.), *Phrase structure and the lexicon*, vol. 33 Studies in natural language and linguistic theory, 109–138. Dordrecht: Kluwer.

Kratzer, Angelika. 2005. Building resultatives. In Claudia Maienborn & Angelika Wöllstein (eds.), *Event arguments: Foundations and applications*, 177–212. Berlin: de Gruyter.

König, Ekkehard. 1991. *The meaning of focus particles: A comparative perspective* Theoretical Linguistics. London and New York: Routledge. doi:10.4324/9780203212288.

Landau, Idan. 2010. The explicit syntax of implicit arguments. *Linguistic Inquiry* 41(3). 357–388.

Landau, Idan. 2013. *Control in Generative Grammar: A Research Companion*. Cambridge University Press.

Legate, Julie Anne. 2003. Some underspecified phases. *Linguistic Inquiry* 34(3). 506–517.

Legate, Julie Anne. 2014. *Voice and v: Lessons from Acehnese* (Linguistic Inquiry Monographs 69). Cambridge, MA: MIT Press.

Lekakou, Marika. 2005. *In the middle, somewhat elevated. the semantics of middles and its cross-linguistic realization*. London, UK: University of London dissertation.

Lenz, Alexandra. 2007. Zur Grammatikalisierung von *geben* im Deutschen und Lëtzebuergeschen. *Zeitschrift für Germanistische Linguistik* 35(1-2). 52–82. doi:10.1515/zgl.2007.004.

Levin, Beth & M. Rappaport Hovav. 2005. *Argument realization* Research Surveys in Linguistics Series. Cambridge, UK: Cambridge University Press.

Levin, Beth & Malka Rappaport Hovav. 1995. *Unaccusativity: At the syntax–lexical semantics interface* (Linguistic Inquiry Monographs 26). Cambridge, MA: MIT Press.

MacDonald, Jonathan E. 2008. *The syntactic nature of inner aspect: A minimalist perspective*, vol. 133. Amsterdam & Philadelphia: John Benjamins Publishing. doi:10.1075/la.133.

Marantz, Alec. 2001. Words. In *The 20th West Coast Conference on Formal Linguistics* (*WCCFL 20*), Somerville: Cascadilla Press.

Marantz, Alec. 2007. Phases and words. In S.-H. Choe (ed.), *Phases in the theory of grammar*, 165–185. Seoul: Dong In.

McIntyre, Andrew. 2004. Event paths, conflation, argument structure and vp shells. *Linguistics* 42(3). 523–571. doi:10.1515/ling.2004.018.

Müller, Gereon. 2014. *Syntactic buffers a local–derivational approach to improper movement, remnant movement, and resumptive movement.*, vol. 91 Linguistische Arbeitsberichte. Leipzig: Universität Leipzig.

Müller, Gereon. 2019. Can unaccusative verbs undergo passivization in German? In J.M.M. Brown, Andreas Schmidt & Marta Wierzba (eds.), *Trees and birds A Festschrift for Gisbert Fanselow*, 135–154. Potsdam: University Press Potsdam. doi:https://doi.org/10.25932/publishup-43225.

Müller, Gereon. 2020. Rethinking restructuring. In András Bárány, Theresa Biberauer, Jamie Douglas & Sten Vikner (eds.), *Syntactic architecture and its consequences ii: Between syntax and morphology*, 149–190. Berlin: Language Science Press. doi:10.5281/zenodo.4280643.

Nübling, Damaris. 2006. Auf Umwegen zum Passivauxiliar - Die Grammatikalisierungspfade von GEBEN, WERDEN, KOMMEN und BLEIBEN im Luxemburgischen, Deutschen und Schwedischen. In Claudine Moulin & Damaris Nübling (eds.), *Perspektiven einer linguistischen Luxemburgistik. Studien zu Diachronie und Synchronie*, 171–201. Heidelberg: Winter.

Petrovičová, Šárka & Svatava Škodová. 2024. Analýza užití slovesa jít s oporou Českého národního korpusu. *Korpus – gramatika – axiologie* 30. 44–64. doi:10.58756/k30241163. [Corpus-based analysis of the usage of the verb *jít* 'to go'].

Pitteroff, Marcel. 2014. *Non-canonical lassen-middles*. Stuttgart: University of Stuttgart dissertation.

Pylkkänen, Liina. 2008. *Introducing arguments* (Linguistic Inquiry Monographs 49). Cambridge, MA: MIT Press.

Ramat, Paolo. 1998. Typological comparison and linguistic areas: some introductory remarks. *Language Sciences* 20(3). 227–240.

Ramchand, Gillian. 2008. *Verb meaning and the lexicon: A first phase syntax* Cambridge Studies in Linguistics. Cambridge: Cambridge University Press.

Rappaport Hovav, Malka. 2008. Lexicalized meaning and the internal temporal structure of events. In Susan Rothstein (ed.), *Theoretical and crosslinguistic approaches to the semantics of aspect*, 13–42. Amsterdam: John Benjamins. doi:10.1075/la.110.03hov.

Rappaport Hovav, Malka & Beth Levin. 2010. Reflections on manner/result complementarity. In Edit Doron, Malka Rappaport Hovav & Ivy Sichel (eds.), *Syntax, lexical semantics, and event structure*, 21–38. Oxford: Oxford University Press.

Reis, Marga. 1979. Ansätze zu einer realistischen grammatik. In Klaus Grubmüller (ed.), *Befund und bedeutung. zum verhältnis von empirie und interpretation in sprach- und literaturwissenschaft*, 1–21. Tübingen: Niemeyer.

Roberts, Ian. 2010. *Agreement and head movement: Clitics, incorporation, and defective goals*. Cambridge, MA: MIT Press.

Roberts, Ian G. 1987. *The representation of implicit and dethematized subjects*. Berlin, New York: De Gruyter Mouton.

Roeper, Thomas. 1987. Implicit arguments and the head-complement relation. *Linguistic Inquiry* 18. 267–310.

Sabel, Joachim. 2001. Deriving multiple head and phrasal movement: The cluster hypothesis. *Linguistic Inquiry* 32(3). 532–547. doi:10.1162/ling.2001.32.3.532.

Schäfer, Florian. 2008. *The syntax of (anti-)causatives*. Amsterdam/Philadelphia: John Benjamins.

Schütze, Carson T. 1996. *The empirical base of linguistics: Grammaticality judgments and linguistic methodology*. Chicago: University of Chicago Press.

Sluckin, Benjamin L. 2021. *Non-canonical subjects and subject positions: locative inversion, V2-violations, and feature inheritance*. Berlin: Humboldt-Universität zu Berlin, Sprach- und literaturwissenschaftliche Fakultät dissertation. doi:http://dx.doi.org/10.18452/23715.

von Stechow, Arnim. 1990. Status government and coherence in German. In Günther Grewendorf & Wolfgang Sternefeld (eds.), *Scrambling and barriers*, vol. 5 Linguistik Aktuell/Linguistics Today, 143–198. John Benjamins. doi:10.1075/la.5.09ste.

von Stechow, Arnim. 1995. Lexical decomposition in syntax. In Urs Egli, Peter E. Pause, Christoph Schwarze, Arnim von Stechow & Götz Wienold (eds.), *Lexical knowledge in the organization of language*, 81–118. Amsterdam: John Benjamins.

Steinbach, Markus. 2002. *Middle voice. a comparative study in the syntax-semantics interface of german* 50. Amsterdam/Philadelphia: John Benjamins. doi:10.1075/la.50.

Sternefeld, Wolfgang. 2006. *Syntax: Eine morphologisch motivierte generative beschreibung des deutschen*. Mannheim: Stauffenburg.

Stroik, Thomas. 1992. Middles and movement. *Linguistic Inquiry* 23(1). 127–138.

Tenny, Carol Lee. 1994. *Aspectual roles and the syntax-semantics interface*. Dordrecht: Kluwer.

Traugott, Elizabeth C. 1993. The conflict promises/threatens to escalate into war. *Berkeley Linguistics Society* (*BLS*) (19). 348–358.

Travis, Lisa deMena. 2010. *Inner aspect: The articulation of the vp* (Studies in Natural Language and Linguistic Theory 80). Dordrecht, Heidelberg, London, New York: Springer. doi:10.1007/978-90-481-8550-4.

Vendler, Zeno. 1957. Verbs and Times. *The Philosophical Review* 66. 143–160.

Vendler, Zeno. 1967. *Linguistics in philosophy*. Ithaca: Cornell University Press.

Vogel, Ralf. 2009. Skandal im verbkomplex. betrachtungen zur scheinbar inkorrekten morphologie in infiniten verbkomplexen des deutschen. *Zeitschrift für Sprachwissenschaft* 28(2). 307–346. doi:10.1515/zfsw.2009.028.

Von Stechow, Armin. 1996. The different readings of *wieder* 'again': A structural account. *Journal of Semantics* 13. 87–138.

Wiemer, Björn. 2011. The grammaticalization of passives. In Heiko Narrog & Bernd Heine (eds.), *The Oxford handbook of grammaticalization*, 535–546. Oxford: Oxford University Press.

Wiesinger, Peter. 1989. Zur Passivbildung mit *kommen* im Südbairischen. In Wolfgang Putschke, Werner Veith & Peter Wiesinger (eds.), *Dialektgeographie und Dialektologie: Günter Bellmann zum 60. Geburtstag von seinen Schülern und Freunden*, 256–268. Marburg: Elwert.

Williams, Edwin. 1987. Implicit arguments, the binding theory, and control. *Natural Language and Linguistic Theory* 5. 151–180.

Wurmbrand, Susi. 1998. *Infinitives*. Cambridge, MA: Massachusetts Institute of Technology dissertation.

Wurmbrand, Susi. 1999. Modal Verbs Must be Raising Verbs. In Sonya Bird, Andrew Carnie, Jason D. Haugen & Peter Norquest (eds.), *Proceedings of WCCFL 18*, 599–612. Somerville, MA: Cascadilla Press.

Wurmbrand, Susi. 2001. *Infinitives: Restructuring and clause structure*. Berlin & New York: Mouton de Gruyter. doi:10.1515/9783110908329.

Wurmbrand, Susi. 2004. Two types of restructuring – lexical vs. functional. *Lingua* 114(8). 991–1014. doi:10.1016/S0024-3841(03)00102-5.

Wurmbrand, Susi. 2015. Complex predicate formation via voice incorporation. In Léa Nash & Pollet Samvelian (eds.), *Approaches to complex predicates*, 248–290. Leiden: Brill.

Wöllstein-Leisten, Angelika. 2001. *Die syntax der dritten konstruktion*. Tübingen: Stauffenburg Verlag.

Benjamin L. Sluckin
Ruhr-Universität Bochum
benjamin.lowellsluckin@rub.de

Eric Fuß
Ruhr-Universität Bochum
eric.fuss@rub.de

AI Acknowledgement:

All theoretical conceptualization, data interpretation, and original prose in this manuscript are solely the authors' work. We used AI models (Gemini Pro 3.1, Claude Cowork/Opus 4.8, and Codex/ChatGPT 5.5) strictly for mechanical and technical tasks: auditing and executing statistical scripts, formatting LaTeX, and surface-level copyediting. We reviewed all AI-processed components and take full responsibility for the final content and any errors.